Learning Temporal Consistency for High Spatial Resolution Remote Sensing Imagery Semantic Change Detection

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***Abstract*— Semantic change detection (SCD) is a crucial task in remote sensing imagery interpretation, which identifies where the changes are and what categories the objects before and after the changes belong to. Postclassification comparison (PCC), as the simplest method, suffers from severe false alarms. While multitask methods face another issue, in that the changed areas in the binary change map have the same object categories in the semantic maps, i.e., semantic change inconsistency. In this study, we leveraged the commutative property of temporal semantic labels in unchanged areas to generate temporal consistency embedding and propose temporal-semantic feature blending (TSFB), which is a feature interaction operation that can dynamically adjust the distance between bitemporal features by controlling a blending weight. In order to realize adaptive temporal consistency learning, we propose the TEmporal-Semantic feature CalibratiOn (TESCO) module, which can estimate the optimal value of the blending weight for TSFB automatically and make the segmentation network learn temporal consistency from coarse to fine through recurrence and the end-to-end training process. The TESCO module is a plug-and-play module that can be combined with any SCD method. In this study, we added the TESCO module to both PCC and multitask-based SCD methods and conducted comprehensive experiments on three large-scale and different application scenario SCD datasets with different semantic change numbers and time series. The experimental results show that the TESCO module can effectively learn temporal consistency, resulting in a significant improvement in the performance of PCC methods and the semantic consistency of the multitask-based methods. The implementation of the TESCO module will be available at https://github.com/Daisy-7/TESCO**

***Index Terms*— Deep learning, remote sensing, semantic change consistency, semantic change detection (SCD), temporal consistency.**

I. Introduction

SEMANTIC change detection (SCD) using high spatial resolution (HSR) remote sensing images has become a meaningful and challenging task in Earth vision. SCD aims to locate where the change occurs while identifying the objects before and after the change. Therefore, SCD takes bitemporal imagery as input and generates pixelwise change and semantic segmentation results for the two images. SCD can provide rich land-cover information for urban planning [1], [2], disaster assessment [3], [4], and environmental protection [5], [6].

The rapid development of deep learning has made it possible to efficiently process the massive data acquired by satellites through long-term observations. Significant progress has been made in tasks such as segmentation, classification, and target detection. Deep learning-based SCD methods are also booming. Intuitively, postclassification comparison (PCC) methods logically compare the segmentation maps from different times generated by a semantic segmentation network to obtain the semantic change result [7]. Since PCC methods separately process bitemporal images, they do not consider the semantic consistency between multitemporal images in unchanged areas, i.e., the temporal consistency, resulting in numerous pseudo-changes in the SCD results, as shown in Fig. 1(a). The existing works consider temporal consistency by introducing binary change detection (BCD) to overcome the problem. The fundamental approach is to decouple SCD as semantic segmentation and BCD [7], [8], [9], [10]. These two tasks can be trained in an end-to-end manner by a multitask framework, as shown in Fig. 1(b), with three outputs: bitemporal segmentation maps and a BCD map. The BCD map masks the bitemporal segmentation maps, obtaining the SCD maps in the inference stage [11]. This postprocessing, which we denote as MASK, is a crucial step for multitask SCD methods to eliminate false positives in PCC since it introduces the BCD result so as to ignore the inconsistent segmentation results in the unchanged areas.

However, this raises another issue: semantic change inconsistency. As shown in Fig. 1(b), the changed area in the BCD map has the same class in the two temporal phases and, therefore, should remain unchanged. Similarly, unchanged areas in the BCD map can have different categories of objects. This phenomenon violates the relationship between object and change, in that the changed area has different objects and the unchanged area has the same objects. The reason why the fundamental multitask SCD methods such as ChangeMask [8] and ChangeStar [9] suffer from semantic change inconsistency is that the temporal consistency is only embedded in the segmentation results by postprocessing in the inference stage, rather

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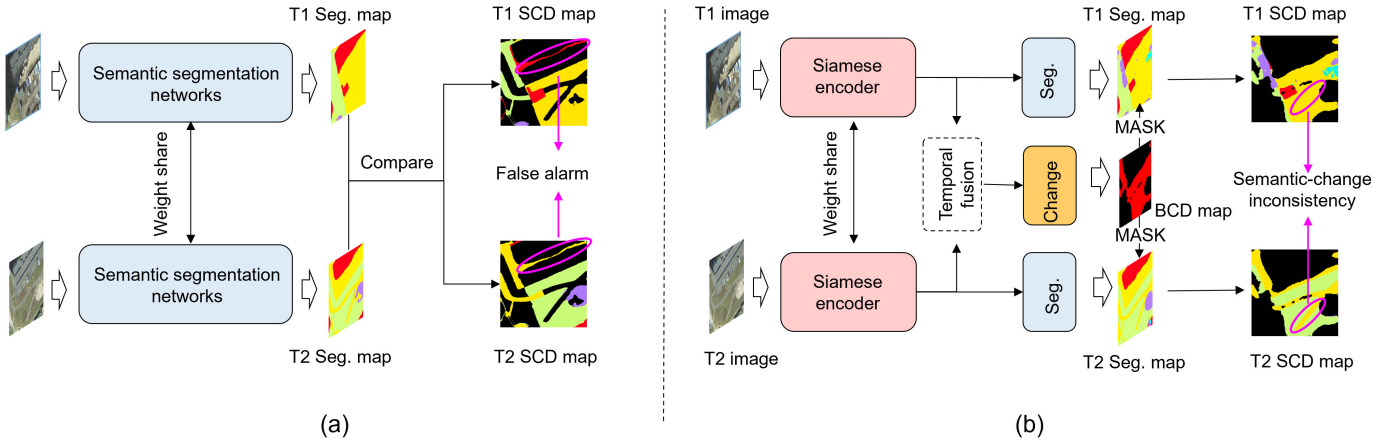


Fig. 1. Framework of two standard SCD methods: (a) PCC and (b) multitask SCD. Temporal fusion can distinguish the change and unchanged areas by aggregating temporal semantic features, e.g., concatenation [9] and absolute differencing [12].

than encoding to the segmentation network. Furthermore, the segmentation networks still cannot learn temporal consistency automatically. Some other methods, such as the change-guided (CG) [13], change-aware (CA) [14] modules, and bitemporal semantic reasoning network (Bi-SRNet) [15], attempt to learn temporal consistency by change feature guidance or similarity loss optimization. Although these methods can improve SCD performance, to some extent, they do not entirely solve the problem of semantic change inconsistency since they lack explicit model temporal consistency between semantic features to constrain the semantic network learning and still need to use MASK to obtain more reasonable results.

Hence, how to robustly model temporal consistency and let the network automatically learn temporal consistency? There are two important characteristics in SCD caused by temporal consistency: 1) the bitemporal semantic labels in unchanged areas have the commutative property and 2) the smaller the distance between bitemporal features in unchanged regions, the more likely they are to be classified into the same category. Based on these characteristics, we encode the temporal consistency into single-temporal semantic features by a blending weight and propose a simple yet effective operation—temporal-semantic feature blending (TSFB)—which calculates the optimal value of the blending weight by minimizing the distance of the temporal semantic features to maintain the semantic consistency of unchanged areas. We also propose a new architectural unit—the Temporal-Semantic feature CalibratiOn (TESCO)—module, which automatically estimates the optimal value of blending weight to embed the temporal consistency into the multitemporal segmentation. The TESCO module can be added to any of the state-of-the-art SCD networks to obtain more consistent and competitive results. The main contributions of this study are summarized as follows.

1) *A Theoretically Interpretable Temporal Consistency Modeling Method Is Proposed:* We propose TSFB to embed temporal consistency into single-temporal semantic features by leveraging the label exchangeability of unchanged regions. We provide mathematical proofs demonstrating that TSFB effectively decreases the feature distance in unchanged regions and preserves the feature distance in changed regions with appropriate blending weights (see Section III-B1).
2) *Adaptive End-to-End Temporal Consistency Learning:* We propose the TESCO module, which is a differentiable computation unit, to adaptively estimate the optimal blending weights for the TSFB and realize cyclic interaction between temporal semantic features to learn temporal consistency from coarse to fine. The ablation studies for different TSFB types and the components of the TESCO module were conducted to investigate the key factors of learning a temporal consistency representation.
3) *Generating More Coherent Results for Both PCC and Multitask SCD Methods:* The TESCO module is light and flexible, which can be easily added to any SCD method and achieve more competitive results with only a slight increase in the parameters and computational complexity. After incorporating the TESCO module into the PCC method, false positives are significantly reduced and the SCD results are comparable to those of the multitask method. When TESCO was incorporated into a multitask approach, its PCC results for the first time exceeded those obtained by binary change mask postprocessing. This makes our method more suitable for continuous SCD on time-series images.

The rest of this article is organized as follows. Section II briefly introduces the deep learning-based SCD methods and the concept of semantic change inconsistency. The details of the proposed module are provided in Section III. Section IV describes the extensive experiments conducted on three different datasets. The ablation study and a detailed discussion are provided in Section V. Section VI concludes this article.

II. Related Work

A. Deep Learning-Based SCD

If two images are in time $t1$ and time $t2$, we denote the number of classes in these two image is n. The semantic change space of these two images is $\vartheta(n*n)$. For SCD, PCC is an intuitive and easy way to detect semantic changes. PCC

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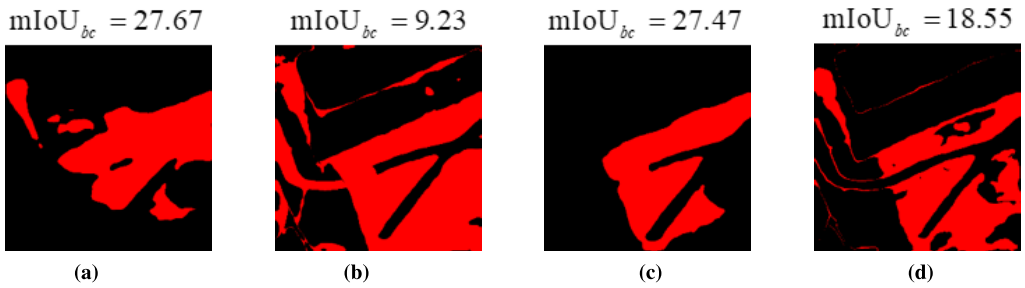


Fig. 2. Performance gap between $O_{\text{bc}}^{\text{net}}$ output from the change detection branch of the multitask SCD network and $O_{\text{bc}}^{\text{pcc}}$ generated by comparing semantic maps. The red area is the changed area, while the black area is unchanged. The semantic change consistency score (mIoU$_{\text{scc}}$) of ChangeStar is only 46.63. After adding the CA module, the mIoU$_{\text{scc}}$ score is 60.44, but semantic change inconsistency still exists. (a) ChangeStar $O_{\text{bc}}^{\text{net}}$. (b) ChangeStar $O_{\text{bc}}^{\text{pcc}}$. (c) + CA $O_{\text{bc}}^{\text{net}}$. (d) + CA $O_{\text{bc}}^{\text{pcc}}$.

simplifies the semantic change space $\vartheta(n*n)$ to the semantic class spaces $\vartheta(n)$, where $\vartheta(n)$ are the semantic class spaces of these two images. The quality of remote sensing semantic segmentation results seriously affects the PCC change detection results. At present, remote sensing semantic segmentation methods are rapidly developing with deep learning [16], [17], [18], [19], [20]. A series of remote sensing semantic segmentation methods based on deep learning has been proposed and gradually applied to challenging scenarios such as cross-city [21], [22], [23], [24] and cross-modal [25], [26], [27]. PCC can directly reuse the state-of-the-art remote sensing segmentation networks based on deep ConvNet [28], [29] or a transformer architecture [30], [31], [32]. However, PCC does not consider temporal consistency when performing semantic segmentation with multitemporal imagery. Therefore, the effect of PCC is affected by different imaging conditions and registration errors of multitemporal imagery. Although the segmentation accuracy of each image can be high, the SCD accuracy for the image pairs drops sharply. Severe false positives in unchanged areas hinder the application of PCC on HSR imagery [7]. In order to reduce the false positives in PCC, BCD can be combined with semantic segmentation to constitute a multitask network [7], which usually has a Siamese encoder to extract temporalwise deep features and two task-specific decoders to output the BCD and segmentation maps [8]. Semantic change maps are generated by MASK, which uses the BCD map to mask the two semantic segmentation maps. This step efficiently avoids false positives in unchanged areas. As a result, the multitask structure has become the mainstream framework of SCD methods, and many works have focused on improving the BCD ability by designing a better temporal fusion module [13], [15], [33], [34]. However, the SCD results typically suffer from semantic change inconsistency [35]. In multitask SCD methods, we denote the BCD map output directly by the BCD network as $O_{\text{bc}}^{\text{net}}$, and the BCD map generated by comparing the segmentation maps of the semantic segmentation network as $O_{\text{bc}}^{\text{pcc}}$. As shown in Fig. 2(a) and (b), we take ChangeStar as an example, and the performance of PCC in a multitask network is largely different with change detection networks. Due to the inconsistency between $O_{\text{bc}}^{\text{net}}$ and $O_{\text{bc}}^{\text{pcc}}$, the semantic change results generated by MASK can have the same ground objects in changed areas and different ground objects in unchanged areas, i.e., semantic change inconsistency [35], as shown in Fig. 1. The fundamental approach to eliminating this inconsistency is to enable semantic feature learning of temporal consistency.

B. Temporal Consistency Learning

Temporal consistency is very important and has been applied in different change detection scenarios. The methods for achieving temporal consistency learning can be divided into two categories.

1) Designing modules to enhance interaction between change and semantic features. For example, the CG module [13] combines change features with semantic features through elementwise addition, while the CA module [14] uses concatenation. The cross-temporal semantic reasoning (Cot-SR) block is proposed in Bi-SRNet [15] to enhance the temporal correlation of semantic features by a cross-attention mechanism. Cui and Jiang [36] proposed a spatial feature enhancement (SFE) module to provide location prior information for semantic features, which helped explore the correlation of the two subtasks. Ding et al. [37] proposed a semantic change transformer (SCanFormer) to detect the "from–to" semantic transitions.
2) Adding constraints in the loss function. For BCD, Hafner et al. [38] design a semi-supervised learning (SSL) loss to predict more consistency results. Similarly, Shu et al. [39] propose a task-level consistency learning loss for semi-supervised building change detection. For SCD, the Bi-SRNet [15] introduces semantic consistency loss (SCLoss) by calculating the similarity between the multitemporal segmentation results. Ding et al. [37] introduce temporal consistency to constrain semantics learning by pseudo-semantic objective loss (PsdLoss). These methods can alleviate the inconsistency slightly, but a significant performance gap still exists between PCC and BCD.

Modeling temporal consistency through cross-attention, transformers, or loss functions does not strictly guarantee a decrease in the distance between semantic features of unchanged regions over time. Consequently, these methods are easily affected by complex land cover distributions, radiometric differences, and registration errors in time-series HSR

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remote sensing imagery. Therefore, it is vital for SCD methods to learn temporal consistency robustly.

III. Approach

A. Temporal Consistency Embedding

Ground objects in unchanged areas should have the same category at different times. Therefore, bitemporal semantic labels in unchanged areas have the commutative property. We utilize this characteristic to generate temporal consistency embedding.

For bitemporal images $\mathbf{X}^{t1}, \mathbf{X}^{t2} \in \mathbb{R}^{W\times H\times C}$, their semantic labels are $\mathbf{Y}^{t1}, \mathbf{Y}^{t2} \in \mathbb{N}^{W\times H\times 1}$. The semantic labels $\mathbf{Y}^{t1}$ and $\mathbf{Y}^{t2}$ can be decoupled as the labels of the change ($\mathbf{Y}_c^{ti}$, where $ti = \{t1, t2\}$) and unchanged areas ($\mathbf{Y}_{\text{nc}}^{ti}$), which can be formulated as follows:

$$\mathbf{Y}^{t1} = \mathbf{Y}_c^{t1} + \mathbf{Y}_{\text{nc}}^{t1}, \quad \mathbf{Y}^{t2} = \mathbf{Y}_c^{t2} + \mathbf{Y}_{\text{nc}}^{t2}. \tag{1}$$

The change labels are $\mathbf{Y}^{t1\to t2} = \{0, 1\}^{W\times H\times 1}$, where 1 represents change and 0 is unchanged. Based on the definition of change, the relationship between $\mathbf{Y}^{t1}$ and $\mathbf{Y}^{t2}$ can be formulated as follows:

$$\begin{cases} \mathbf{Y}_c^{t1} \neq \mathbf{Y}_c^{t2}, & \text{changed area} \\ \mathbf{Y}_{\text{nc}}^{t1} = \mathbf{Y}_{\text{nc}}^{t2}, & \text{unchanged area} \end{cases} \tag{2}$$

and

$$\begin{aligned} \mathbf{Y}_c^{ti} &= \mathbf{Y}^{ti} \odot \mathbf{Y}^{t1\to t2} \\ \mathbf{Y}_{\text{nc}}^{ti} &= \mathbf{Y}_{\text{nc}}^{ti} \odot (1 - \mathbf{Y}^{t1\to t2}) \end{aligned} \tag{3}$$

where $\odot$ represents the Hadamard product. Therefore, we can combine (1)–(3) to obtain a new equation

$$\begin{aligned} \mathbf{Y}^{t1} &= \mathbf{Y}_c^{t1} + \mathbf{Y}_{\text{nc}}^{t1} = \mathbf{Y}_c^{t1} + \mathbf{Y}_{\text{nc}}^{t2} \\ &= \mathbf{Y}^{t1} \odot \mathbf{Y}^{t1\to t2} + \mathbf{Y}^{t2} \odot (1 - \mathbf{Y}^{t1\to t2}) \\ \mathbf{Y}^{t2} &= \mathbf{Y}_c^{t2} + \mathbf{Y}_{\text{nc}}^{t2} = \mathbf{Y}_c^{t2} + \mathbf{Y}_{\text{nc}}^{t1} \\ &= \mathbf{Y}^{t2} \odot \mathbf{Y}^{t1\to t2} + \mathbf{Y}^{t1} \odot (1 - \mathbf{Y}^{t1\to t2}). \end{aligned} \tag{4}$$

Equation (4) is a new form of the single-temporal semantic labels, which are identical to the original semantic labels. Equation (4) leverages the commutative property of semantic labels in the unchanged areas to explicitly encode the temporal consistency between bitemporal semantic labels into single-temporal semantic labels. Therefore, we generate temporal consistency embedding by exchanging temporal features. The temporal semantic features extracted by the semantic network are denoted as F_{t1}, $F_{t2} \in \mathbb{R}^{W\times H\times C}$. Due to there being no change labels at the feature level, we introduce a blending weight $W_b \in [0, 1]$ with shape $W \times H \times 1$. Therefore, (4) can be rewritten as follows:

$$\begin{aligned} F'_{t1} &= F_{t1} \odot W_b + F_{t2} \odot (1 - W_b) \\ F'_{t2} &= F_{t2} \odot W_b + F_{t1} \odot (1 - W_b) \end{aligned} \tag{5}$$

where F'_{t1} and F'_{t2} represent the temporal consistency embeddings. The value of W_b is crucial because it determines whether temporal consistency is contained in the temporal features. Therefore, we propose TSFB, including three approaches of bitemporal feature blending. For each blending approach, it has an optimal value of W_b in the change and unchanged areas, to maintain the temporal consistency between features. To automatically estimate W_b, we propose the TESCO module, which can allow the SCD network to learn temporal consistency by end-to-end training.

B. Temporal-Semantic Feature Calibration

As shown in Fig. 3, the TESCO module is a computational unit consisting of blending weight estimation, TSFB, and spatiotemporal feature update. As TSFB influences the way of estimating the blending weight W_b, we first introduce TSFB.

1) Temporal-Semantic Feature Blending: The smaller the feature distance of the unchanged ground objects, the more likely they are to be classified into the same class. Therefore, TSFB aims to minimize the distance of the bitemporal semantic features to maintain semantic consistency in unchanged areas and keep discriminative features in the changed areas, to obtain better SCD results. Based on (5), there are three ways to blend bitemporal semantic features. Examples of the three blending approaches are shown in Fig. 4.

a) Single direction T1 → T2: Use the semantic features of T1 to blend the semantic features of T2. Therefore, the new temporal semantic features F'_{t1} and F'_{t2} are obtained

$$\begin{aligned} F'_{t1} &= F_{t1} \\ F'_{t2} &= F_{t2} \odot W_b + F_{t1} \odot (1 - W_b). \end{aligned} \tag{6}$$

The feature distance of F'_{t1} and F'_{t2} can be calculated as follows:

$$\begin{aligned} |F'_{t1} - F'_{t2}| &= |F_{t1} - F_{t2} \odot W_b - F_{t1} \odot (1 - W_b)| \\ &= |F_{t1} \odot W_b - F_{t2} \odot W_b| \\ &= W_b \odot |F_{t1} - F_{t2}|. \end{aligned} \tag{7}$$

When $W_b = [0, 1)$ in unchanged areas, the distance of the temporal semantic features is minimized. The feature distance is not changed when $W_b = 1$ in the changed areas.

b) Single direction T2 → T1: Use the semantic features of T2 to blend the semantic features of T1. Therefore, the new temporal semantic features F'_{t1} and F'_{t2} are obtained

$$\begin{aligned} F'_{t1} &= F_{t1} \odot W_b + F_{t2} \odot (1 - W_b) \\ F'_{t2} &= F_{t2}. \end{aligned} \tag{8}$$

The feature distance is identical to T1 → T2, i.e., $|F'_{t1} - F'_{t2}| = W_b * |F_{t1} - F_{t2}|$. The ideal value of W_b in the unchanged area is 0, and the ideal value of W_b in the changed area is 1.

c) Bidirectional T1 ↔ T2: The semantic features of T1 and T2 are blended with each other. Therefore, the new temporal semantic features F'_{t1} and F'_{t2} are obtained

$$\begin{aligned} F'_{t1} &= F_{t1} \odot W_b + F_{t2} \odot (1 - W_b) \\ F'_{t2} &= F_{t2} \odot W_b + F_{t1} \odot (1 - W_b). \end{aligned} \tag{9}$$

The feature distance of F'_{t1} and F'_{t2} can be calculated as follows:

$$\begin{aligned} |F'_{t2} - F'_{t1}| &= |F_{t2} \odot W_b + F_{t1} \odot (1 - W_b) \\ &\quad - F_{t1} \odot W_b - F_{t2} \odot (1 - W_b)| \\ &= |F_{t2} \odot (2W_b - 1) - F_{t1} \odot (2W_b - 1)| \\ &= |2W_b - 1| \odot |F_{t2} - F_{t1}|. \end{aligned} \tag{10}$$

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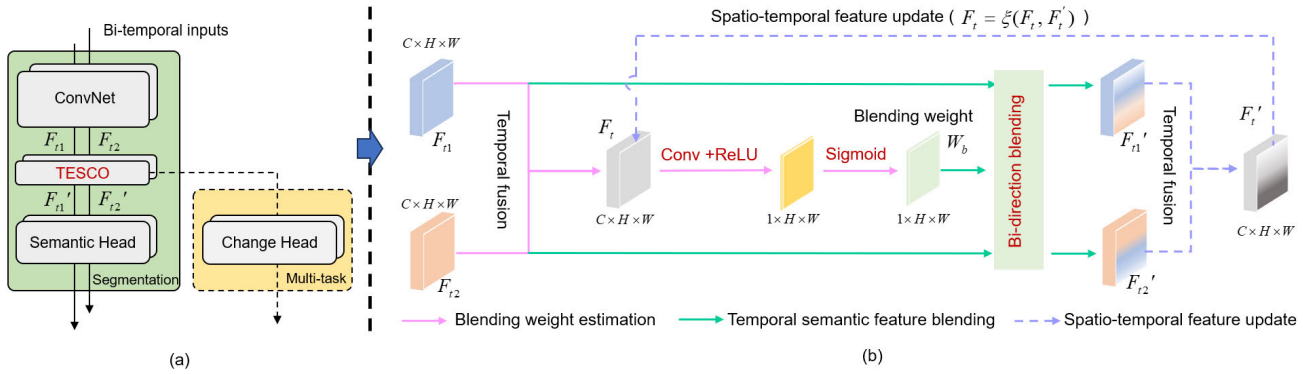


Fig. 3. Flowchart of the TESCO module, which includes three parts: 1) blending weight estimation; 2) TSFB; and 3) spatiotemporal feature update. The dashed line part of (a) is only used for training with multitask supervision. (a) PCC or multitask methods with TESCO. (b) Flowchart of the TESCO module.

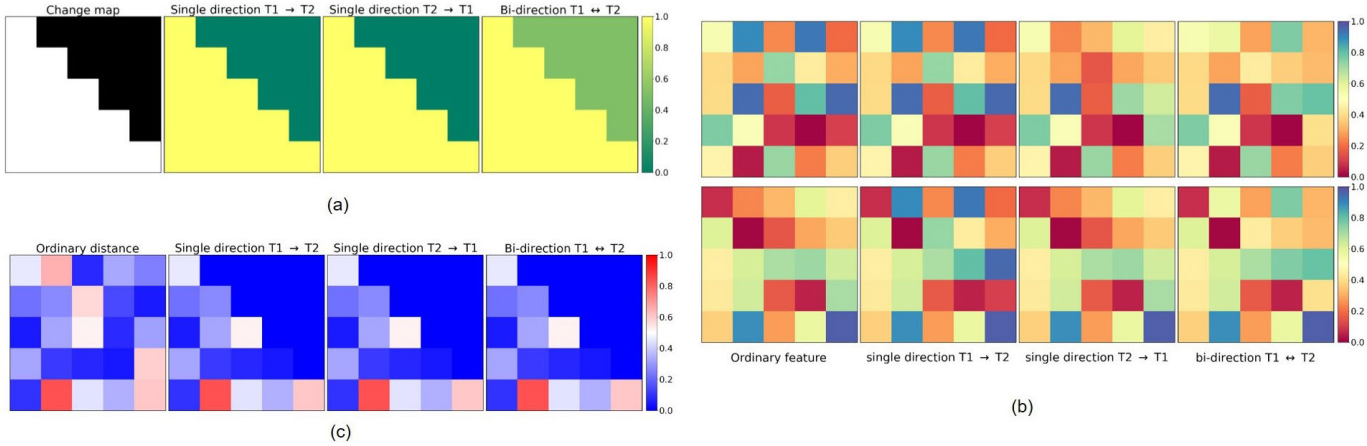


Fig. 4. Examples of the three different blending cases described in Section III-B1. The ordinary bitemporal semantic features and one change map are randomly generated. The white represents the changed pixels for the change map, and the black represents unchanged. After blending, the feature distances of the changed areas remain constant and the feature distances of the unchanged areas decrease to zero. (a) Ideal value of W_b in different blending ways and regions. (b) Temporal semantic features before and after blending. The first row is F_{t1} and the second row is F_{t2}. (c) Feature distance for temporal semantic features before and after blending.

When W_b is in the range $(0, 1)$, the feature distance will be decreased, and it reduces to 0 when $W_b = 0.5$. The discrimination of different temporal semantic features can be maintained when $W_b = 1$ or 0.

2) Blending Weight Estimation: In order to automatically estimate the value of W_b in the changed and unchanged area, temporal fusion δ is first used to generate spatiotemporal features $F_t = \delta(F_{t1}, F_{t2})$, which can distinguish the changed and unchanged areas by aggregating the temporal semantic features. The commonly used temporal fusion operations include concatenation and absolute differencing [7], [40]. A 3×3 convolutional layer is then used to reduce the channel number of F_t to 1, obtaining the initial W_b. The value range of the initial W_b is $(-\infty, +\infty)$.

For single-direction blending, the sigmoid function σ restricts $W_b \in [0, 1]$, which makes the value of W_b in the unchanged areas close to 0 and in changed areas close to 1. For bidirectional blending, a rectified linear unit (ReLU) is added before the sigmoid function, which changes the value range from $(-\infty, +\infty)$ to $[0, +\infty)$. Finally, the value range of W_b is $[0.5, 1]$ when using bidirectional blending, which effectively ensures that the value of W_b in unchanged areas is close to 0.5, and the value in changed areas is close to 1. This is beneficial for maintaining semantic consistency in unchanged areas

$$W_b = \begin{cases} \sigma(\text{Conv}(F_t)), & \text{single direction} \\ \sigma(\text{ReLU}((\text{Conv}(F_t)))), & \text{bidirectional.} \end{cases} \tag{11}$$

3) Spatiotemporal Feature Update: After blending the temporal semantic features, the new temporal semantic features F'_{t1} and F'_{t2} are obtained. Temporal fusion is then used to obtain the new spatiotemporal features $F'_t = \delta(F'_{t1}, F'_{t2})$, which are then used to update F_t for the recurrent TESCO module, improving temporal consistency of temporal semantic feature from coarse to fine

$$F_t = \xi\left(F_t, F'_t\right) \tag{12}$$

where ξ denotes the update methods. In this article, we compare four update methods, including: 1) identity, $\xi = F'_t$; 2) addition, $\xi = F_t + F'_t$; 3) averaging, $\xi = (F_t + F'_t)/2$; and 4) a convolutional layer after concatenation (CatConv), $\xi = \text{CatConv}(F'_t, F_t)$.

C. Comparison of SCD Architectures

The TESCO module can be easily combined with PCC or a multitask method to construct a new architecture, called

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Algorithm 1 Adding the TESCO Module to a PCC Method

Input: Bi-temporal images $\mathbf{X}^{t1}$ and $\mathbf{X}^{t2}$ of size $W \times H \times C$
Output: Bi-temporal semantic maps O_{seg}^{t1}, O_{seg}^{t2}, BCD map O_{bc}^{pcc}, SCD maps O_{sc}^{t1}, O_{sc}^{t2}

Initialize ConvNet θ, semantic head $\mathcal{H}$, temporal fusion δ, spatio-temporal fusion ξ, recurrence of the TESCO module R, $i = 0$ ▷ Supervised by semantic segmentation
Obtain temporal semantic feature $F_{t1} = \theta(\mathbf{X}^{t1})$, $F_{t2} = \theta(\mathbf{X}^{t2})$
Generate spatial-temporal feature $F_t' = \delta(F_{t1}, F_{t2})$ ▷ TESCO module beginning
while $i < R$ **do**
    Estimate blending weight: $W_b = \sigma(\mathrm{ReLU}((\mathrm{conv}(F_t))))$
    Blending temporal semantic features: $F_{t1}' = F_{t1} * W_b + F_{t2} * (1 - W_b)$,
    $F_{t2}' = F_{t2} * W_b + F_{t1} * (1 - W_b)$
    if $i > 0$ **then**
        Update spatial-temporal features: $F_t = \xi(\delta(F_{t1}', F_{t2}'), F_t)$
    end if
    Update temporal semantic features: $F_{t1} = F_{t1}'$, $F_{t2} = F_{t2}'$
    $i = i + 1$
end while ▷ TESCO module end
Generate bi-temporal semantic segmentation map: $\mathrm{O}_{seg}^{t1} = \mathcal{H}(F_{t1})$, $\mathrm{O}_{seg}^{t2} = \mathcal{H}(F_{t2})$
Generate binary change maps: $\mathrm{O}_{bc}^{pcc} = \mathrm{O}_{seg}^{t1} \oplus \mathrm{O}_{seg}^{t2}$ ▷ Inference stage
Generate semantic change maps: $\mathrm{O}_{sc}^{t1} = \mathrm{O}_{seg}^{1} \otimes \mathrm{O}_{bc}^{pcc}$, $\mathrm{O}_{sc}^{t2} = \mathrm{O}_{seg}^{t2} \otimes \mathrm{O}_{bc}^{pcc}$

Algorithm 2 Adding the TESCO Module to a Multitask Method

Input: Bi-temporal images $\mathbf{X}^{t1}$ and $\mathbf{X}^{t2}$ of size $W \times H \times C$
Output: Bi-temporal semantic maps O_{seg}^{t1}, O_{seg}^{t2}, BCD map O_{bc}^{pcc}, SCD maps O_{sc}^{t1}, O_{sc}^{t2}

Initialize ConvNet θ, semantic head $\mathcal{H}$, change head $\mathcal{K}$, temporal fusion δ, spatio-temporal fusion ξ, recurrence of the TESCO module R, $i = 0$ ▷ Supervised by multi-task
Obtain temporal semantic feature $F_{t1} = \theta(X_1)$, $F_{t2} = \theta(X_2)$
Generate spatial-temporal feature $F_t = \delta(F_{t1}, F_{t2})$
while $i < R$ **do** ▷ TESCO module beginning
    Estimate blending weight: $W_b = \sigma(\mathrm{ReLU}(\mathrm{conv}(F_t)))$
    Blending temporal semantic features: $F_{t1}' = F_{t1} * W_b + F_{t2} * (1 - W_b)$,
    $F_{t2}' = F_{t2} * W_b + F_{t1} * (1 - W_b)$
    Update temporal semantic features: $F_{t1} = F_{t1}'$, $F_{t2} = F_{t2}'$
    Update spatial-temporal features : $F_t = \xi(\delta(F_{t1}', F_{t2}'), F_t)$
    $i = i + 1$
end while ▷ TESCO module end
Generate BCD map: $\mathrm{O}_{bc}^{net} = \mathcal{K}(F_t)$
Generate bi-temporal semantic maps: $\mathrm{O}_{seg}^{t1} = \mathcal{H}(F_{t1})$, $\mathrm{O}_{seg}^{t2} = \mathcal{H}(F_{t2})$
Generate BCD map: $\mathrm{O}_{bc}^{pcc} = \mathrm{O}_{seg}^{t1} \oplus \mathrm{O}_{seg}^{t2}$ ▷ Inference stage
Generate SCD maps: $\mathrm{O}_{sc}^{t1} = \mathrm{O}_{seg}^{1} \otimes \mathrm{O}_{bc}^{pcc}$, $\mathrm{O}_{sc}^{t2} = \mathrm{O}_{seg}^{t2} \otimes \mathrm{O}_{bc}^{pcc}$

ChangeSeg. Generally speaking, the TESCO module is added before the head of any SCD network, where the features have distinguishable semantic information. In order to reuse the temporal fusion of SCD methods, the TESCO module can also be added where the temporal fusion takes place. The pseudocodes of the TESCO module in different SCD frameworks are shown in Algorithms 1 and 2.

We compared ChangeSeg with other SCD architectures to show their differences. As shown in Fig. 5, the PCC architecture lacks a temporal consistency learning module in the segmentation network and has difficulty in overcoming the influence of radiometric differences and registration errors in HSR multitemporal images, resulting in severe false alarms. Although the multitask architectures, such as ChangeStar [as shown in Fig. 5(b)], consider the temporal correlation of bitemporal images by temporal fusion to detect the binary change and use MASK to generate SCD results, they do not consider the temporal consistency in the segmentation network, resulting in semantic change inconsistency. Thanks to the TESCO module, the ChangeSeg architecture can easily embed temporal consistency into the segmentation model and ensure temporal consistency between bitemporal features, achieving competitive results through logically comparing the bitemporal semantic maps, whether supervised by segmentation or multitask supervision.

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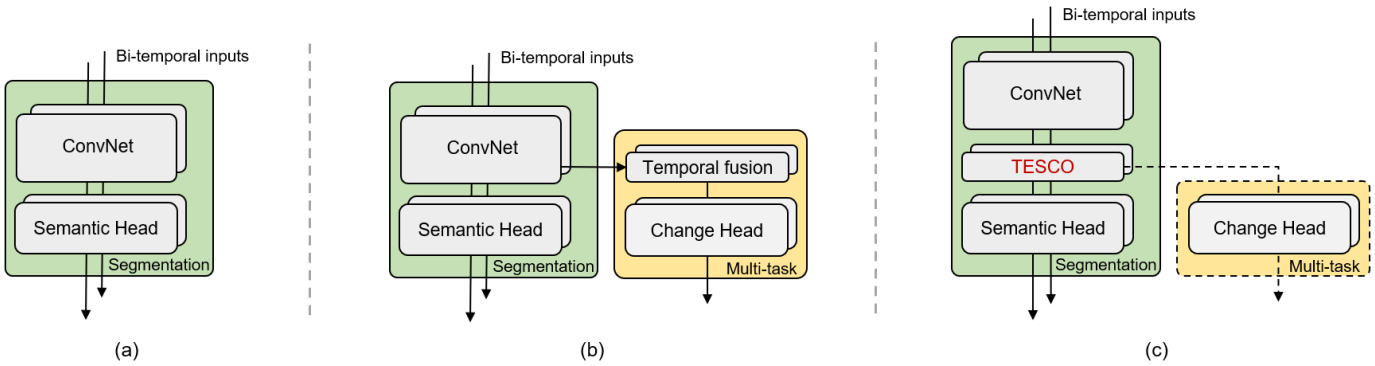


Fig. 5. Comparison of SCD architectures. ChangeSeg embeds the temporal correlation into the segmentation network by the TESCO module with segmentation or multitask supervision. In the inference stage, the results of PCC and ChangeSeg are generated by comparing the bitemporal segmentation results. ChangeStar uses the binary change map output by the change network to mask the bitemporal segmentation results and to obtain the semantic change results. The dashed line part of ChangeSeg is only used for training with multitask supervision. (a) PCC. (b) ChangeStar. (c) ChangeSeg.

TABLE I
DETAILS OF SCD BASELINE, INCLUDING MULTITASK AND PCC METHODS

Multi-task Methods	Segmentation model	Ordinary temporal fusion	Where to add the TESCO module
ChangeMask [8]	U-Net [42]	TST (middle fusion)	(1) Encoder; (2) before head
ChangeStar [9]	FarSeg [41]	CatConv (late fusion)	Before head

IV. EXPERIMENTS

A. Benchmark Methods

1) Baseline: To evaluate the effectiveness and robustness of the TESCO module, we selected two state-of-the-art multitask SCD methods and their associated segmentation models as the baseline, as shown in Table I. ChangeMask is a multitask network based on U-Net, which has a Siamese encoder to extract semantic features and then task-specific decoders to recover the semantic information and change information fused by the temporal-symmetric transformer (TST). ChangeStar consists of a deep segmentation model (FarSeg [41]) to segment the images and a convolutional module (CatConv), which is followed by a concatenation operation to detect change. The baseline, respectively, represents middle and late temporal fusion in the multitask SCD methods.

2) Adding the TESCO Module to SCD Methods: For middle fusion SCD methods such as ChangeMask, the ordinary temporal fusion operation was replaced by the TESCO module, and the TESCO module was also added before the semantic head. For the late fusion SCD methods such as ChangeStar, we only replaced the ordinary temporal fusion operation with the TESCO module. The network of the PCC methods is the segmentation model of the corresponding multitask methods, so the TESCO module is added at the same location. The temporal fusion operation in the TESCO module was added at the same location. The temporal fusion operation in the TESCO module was the same as the baseline, for a fair comparison. After adding the TESCO module to the multitask methods, the SCD results were generated by PCC. In contrast, the SCD results of the other multitask supervised methods were obtained by MASK. Unless otherwise specified, the recurrence of the TESCO module was 1, and $\xi = F_t + F_t'$. In all experiments, the multitask methods were supervised by both the BCD task and semantic segmentation task. We adopted binary cross-entropy loss and dice loss to supervise the BCD task. The semantic segmentation task was supervised by cross-entropy loss.

B. Metrics

In reference to [35], we use variations of the intersection over union (IoU) to assess the performance of the TESCO module in semantic segmentation, BCD, and SCD.

1) Semantic Segmentation: The performance of semantic segmentation is evaluated by $\mathrm{mIoU}_{\mathrm{seg}} = (\mathrm{mIoU}_{\mathrm{seg}}^{2018} + \mathrm{mIoU}_{\mathrm{seg}}^{2019})/2$, where $\mathrm{mIoU}_{\mathrm{seg}}^{2018}$ and $\mathrm{mIoU}_{\mathrm{seg}}^{2019}$ represent the performance in 2018 and 2019, respectively.

2) Binary Change Detection: Only the IoU of the positive class (changed), i.e., $\mathrm{IoU}_{\mathrm{bc}}$ to assess the BCD performance.

3) Semantic Change Detection: The performance of the changed and unchanged areas is evaluated separately. The number of semantic transitions in changed areas is denoted as n_c and the number of semantic transitions in unchanged areas is denoted as n_{nc}. For changed areas, $\mathrm{mIoU}_{\mathrm{sc}}^{c} = (1/(n_c+1))(\sum_{i=1}^{n_c} \mathrm{IoU}_i + \mathrm{IoU}_{\mathrm{nc}})$, where $\mathrm{IoU}_{\mathrm{nc}}$ is the IoU of unchanged areas in the binary change result. For unchanged areas, $\mathrm{mIoU}_{\mathrm{sc}}^{\mathrm{nc}} = (1/(n_{\mathrm{nc}}+1))(\sum_{i=1}^{n_{\mathrm{nc}}} \mathrm{IoU}_i + \mathrm{IoU}_{\mathrm{bc}})$. The $\mathrm{mIoU}_{\mathrm{sc}} = (\mathrm{mIoU}_{\mathrm{sc}}^{c} + \mathrm{mIoU}_{\mathrm{sc}}^{\mathrm{nc}})/2$ is calculated as a final metric for SCD.

4) Semantic Change Consistency: Especially for multitask SCD methods, the semantic change consistency is evaluated by calculating the IoU for the changed and unchanged areas between the BCD map directly output by the network and the BCD map obtained by comparing the semantic segmentation results for different years. The $\mathrm{mIoU}_{\mathrm{scc}} = (\mathrm{IoU}_{\mathrm{scc}}^{c} + \mathrm{IoU}_{\mathrm{scc}}^{\mathrm{nc}})/2$,

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TABLE II
QUANTITATIVE ASSESSMENT RESULTS OF SCD BASELINES WITH/WITHOUT THE TESCO MODULE ON THE HI-UCD TEST SET

Method	Backbone	Semantic segmentation			Semantic change detection			Binary change detection	Params (M)	MAdds (B)
		$mIoU_{seg}(\%)$	$mIoU^{2018}_{seg}(\%)$	$mIoU^{2019}_{seg}(\%)$	$mIoU_{sc}(\%)$	$mIoU^{c}_{sc}(\%)$	$mIoU^{nc}_{sc}(\%)$	$IoU_{bc}(\%)$		
Multi-task supervised										
ChangeMask [8]	EfficientNet-B0	**56.12**	55.95	56.28	**33.97**	14.49	53.44	**32.15**	10.62	9.48
+ TESCO		**57.42**(+1.30)	57.50	57.35	**35.24**(+1.27)	16.47	54.01	**33.48**(+1.33)	10.63(+0.01)	11.26(+1.78)
ChangeStar [9]	ResNet-34	**56.31**	55.56	57.05	**31.92**	12.73	51.12	**27.67**	26.51	21.18
+ TESCO		**57.60** (+1.29)	57.57	57.62	**34.28**(+2.26)	14.53	54.04	**28.84**(+1.17)	26.51(+0.00)	21.87(+0.69)
Segmentation supervised										
U-Net [42]	EfficientNet-B0	**58.79**	58.58	58.99	**31.62**	9.46	53.78	**8.74**	6.25	5.95
+ TESCO		**59.03**(+0.24)	59.32	58.75	**35.31**(+3.69)	14.73	55.89	**30.29**(+21.55)	8.39(+2.14)	7.68(+0.73)
FarSeg [41]	ResNet-34	**60.36**	60.00	60.72	**31.43**	9.90	52.96	**8.54**	26.34	20.50
+ TESCO		**61.16**(+0.80)	60.96	61.37	**36.44**(+5.01)	15.81	57.07	**26.38**(+17.84)	26.51(+0.21)	21.19(+0.69)

where IoU^{c}_{scc} and IoU^{nc}_{scc} represent the IoU in changed and unchanged areas, respectively.

C. Land-Use and Land-Cover SCD

We conducted experiments on the Hi-UCD dataset, evaluating the effectiveness of the TESCO module in land-use and land-cover (LULC) SCD task with bitemporal images.

1) Hi-UCD Dataset: The Hi-UCD dataset is made up of a total of 40 800 pairs of 512 × 512 patches with a very high resolution of 0.1 m, of which 12 000 pairs form the training set, 7200 pairs form the validation set, and the remaining 21 600 pairs form the test set. There are 48 types of semantic change (except for unchanged) in the whole dataset, of which 31 types are in the validation set and 39 are in the test set. The extensive data volume, severe sample imbalance of semantic changes, and refined semantic annotation make the Hi-UCD dataset more challenging for the SCD task.

2) Implementation Details: All the models were trained for 50k iterations with a poly learning rate policy, where the initial learning rate was set to 0.01 and multiplied by $(1 - (\text{step}/\max \text{step}))^{\gamma}$ with $\gamma = 0.9$. We used stochastic gradient descent (SGD) as the optimizer on a single Titan RTX GPU with a weight decay of 0.0001 and a momentum of 0.9. The batch size was 16. We used horizontal and vertical flips, and rotation of $90{*}k$ $(k = 1, 2, 3)^{\circ}$ as training data augmentation. The number of operations measured by multiply-adds (MAdds), which represents the computational complexity of a network, was estimated for an input of $1 \times 6 \times 256 \times 256$. All results were performed on the Hi-UCD test set.

3) Main Results: The qualitative assessment results of combining the TESCO module with different SCD methods are listed in Table II. Regardless of the form of supervision, all the metrics are significantly improved after adding the TESCO module. For the multitask methods with the TESCO module, the accuracy of the SCD results generated by PCC exceeds those generated by MASK for the first time. This is because the TESCO module increases the interaction between the semantic segmentation and change detection tasks, improving the multitask learning effectiveness. Compared with the PCC-based methods, the greater increase of $mIoU_{seg}$ in the multitask methods illustrates that the TESCO module can alleviate the negative transfer of semantic segmentation when training with BCD. The significant improvement of BCD and SCD for the PCC methods confirms that temporal consistency is important for segmentation in SCD, whereas the TESCO module can successfully help the segmentation network to automatically learn temporal consistency, and thus predicate more consistent semantic results.

In Fig. 6, we first visualize the segmentation results of FarSeg with the TESCO module on the whole Hi-UCD test set, which obtains the highest semantic segmentation score. These two segmentation maps [Fig. 6(a) and (b)] are more consistent in unchanged areas. In order to show the details, we visualize the baseline results with/without the TESCO module in changed and unchanged areas. After adding the TESCO module to the multitask methods, the accuracy and completeness of the changed areas are improved. Since there is a lack of supervision of BCD, the TESCO module also reduces the distance of semantic features in the changed areas, resulting in missed detection in the results of PCC methods. Overall, the use of the TESCO module eliminates the false alarms of the PCC methods and the semantic change inconsistency of the multitask methods.

D. Object SCD

Certain ground objects, such as buildings, are important in Earth vision and are crucial information for urban development. Detecting the semantic change of a single type of object of interest can provide rich information for the downstream task. In this section, we describe how we detected the semantic change of buildings, i.e., new construction and demolition of buildings, to further assess the robustness of the TESCO module with small numbers of semantic change types.

1) SpaceNet 7 Dataset [43]: The SpaceNet 7 dataset is a building change detection dataset, whose spatial resolution is 4 m. The released SpaceNet 7 dataset includes images and building footprints between 2017 and 2020 in 60 regions of interest (ROIs) across the globe. Each ROI has approximately 24 images (one per month) with the size of 1024 × 1024 pixels. We randomly selected 50 ROIs for training, five ROIs

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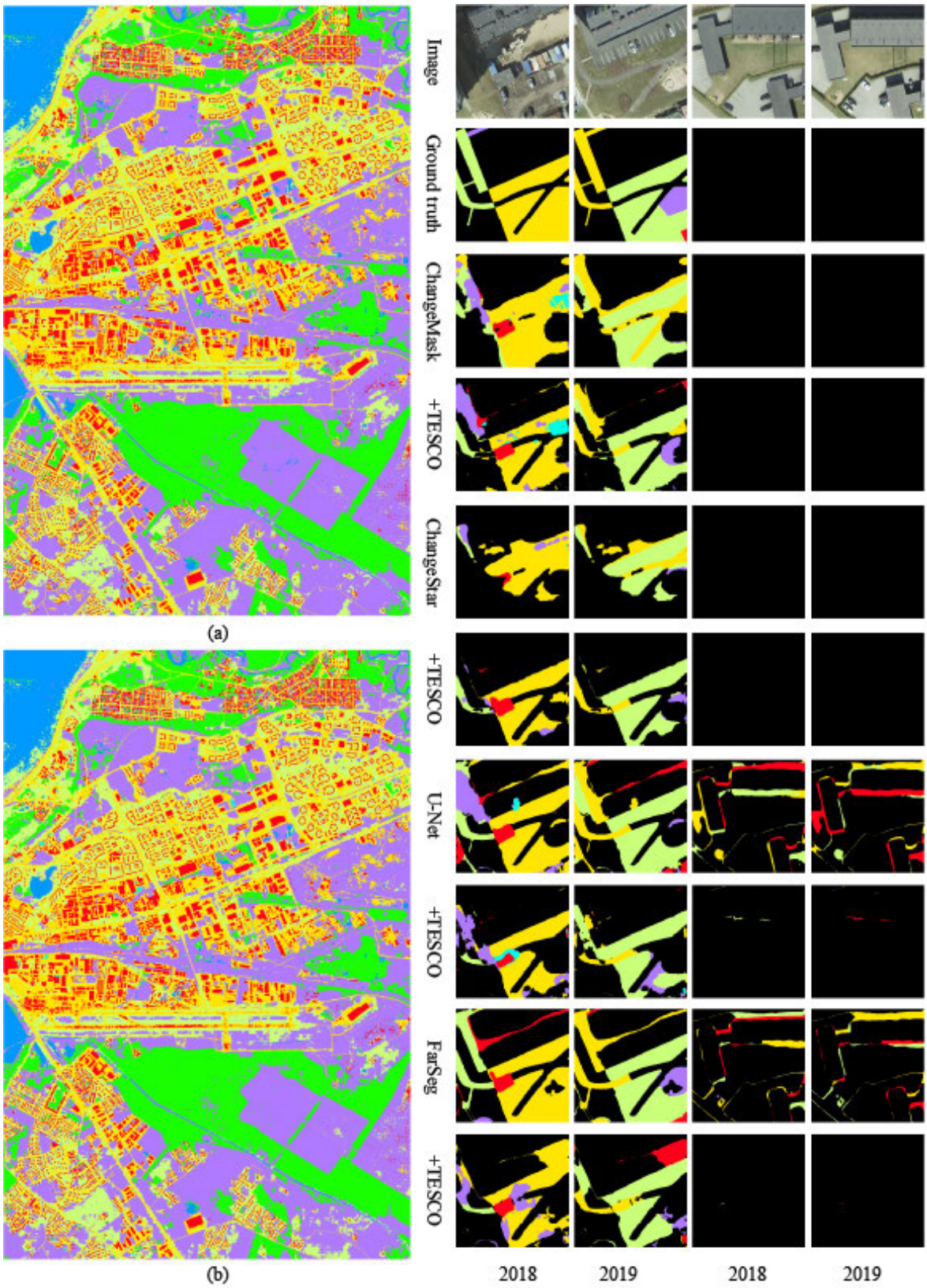


Fig. 6. Visualization results of the SCD baseline with/without the TESCO module on the Hi-UCD test set dataset. Legend: water, grass, building, greenhouse, road, bridge, bare land, woodland, other, unchanged, and white is unlabeled. (a) 2018 segementation map (FarSeg + TESCO). (b) 2019 segementation map (FarSeg + TESCO).

for validation, and five ROIs for testing. For each ROI, every monthly image was combined with a randomly selected image to construct bitemporal image pairs. Therefore, there were approximately 24 pairs of images in each ROI.

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TABLE III
QUANTITATIVE ASSESSMENT RESULTS OF SCD BASELINES WITH/WITHOUT THE TESCO MODULE ON THE SPACENET 7 TEST SET

Method	Backbone	Semantic segmentation			Semantic change detection			Binary change detection
		$\text{mIoU}_{\text{seg}}(\%)$	$\text{mIoU}^{t1}_{\text{seg}}(\%)$	$\text{mIoU}^{t2}_{\text{seg}}(\%)$	$\text{mIoU}_{\text{sc}}(\%)$	$\text{mIoU}^{\text{c}}_{\text{sc}}(\%)$	$\text{mIoU}^{\text{nc}}_{\text{sc}}(\%)$	$\text{IoU}_{\text{bc}}(\%)$
Multi-task supervised								
ChangeMask [8]	EfficientNet-B0	**42.34**	42.34	42.35	**52.71**	51.19	54.23	**27.11**
+ TESCO		**44.42**$_{(+2.08)}$	44.38	44.47	**54.14**$_{(+1.43)}$	51.93	56.35	**28.32**$_{(+1.21)}$
ChangeStar [9]	ResNet-34	**36.53**	36.54	36.53	**50.06**	49.21	50.90	**24.04**
+ TESCO		**37.56**$_{(+1:03)}$	37.53	37.60	**51.61**$_{(+0.71)}$	50.26	52.96	**25.70**$_{(+1.66)}$
Segmentation supervised								
U-Net [42]	EfficientNet-B0	**41.64**	41.63	41.65	**33.82**	34.70	32.94	**8.79**
+ TESCO		**43.97**$_{(+2:33)}$	43.92	44.01	**52.44**$_{(+18.62)}$	49.82	55.07	**25.13**$_{(+16.34)}$
FarSeg [41]	ResNet-34	**35.67**	35.67	35.67	**41.65**	38.03	45.28	**8.64**
+ TESCO		**37.21**$_{(+1:54)}$	37.17	37.24	**51.15**$_{(+9.50)}$	49.72	52.57	**24.89**$_{(+16.25)}$

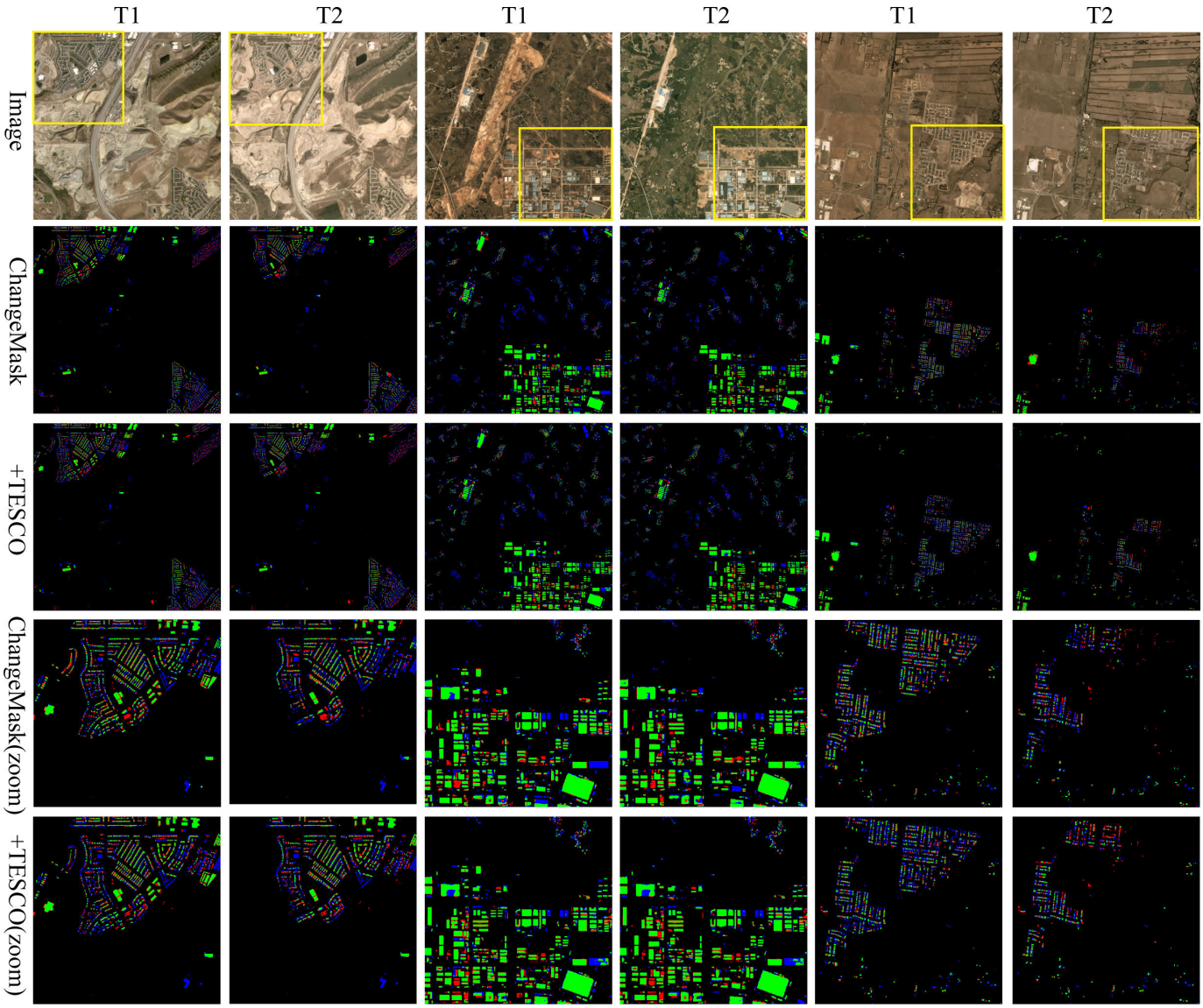


Fig. 7. Visualization of the semantic segmentation results of the ChangeMask with/without the TESCO module on the SpaceNet 7 test set. The fourth and fifth rows are the zoomed-in view results of the yellow box in the images. Legend: ■ true positive, ■ false positive, ■ false negative, and ■ no building.

2) Implementation Details: All the methods were trained for 25k iterations with a poly learning rate policy, where the initial learning rate was set to 0.01 and multiplied by $(1 - (\text{step}/\text{max step}))^{\gamma}$ with $\gamma = 0.9$. We used SGD as the

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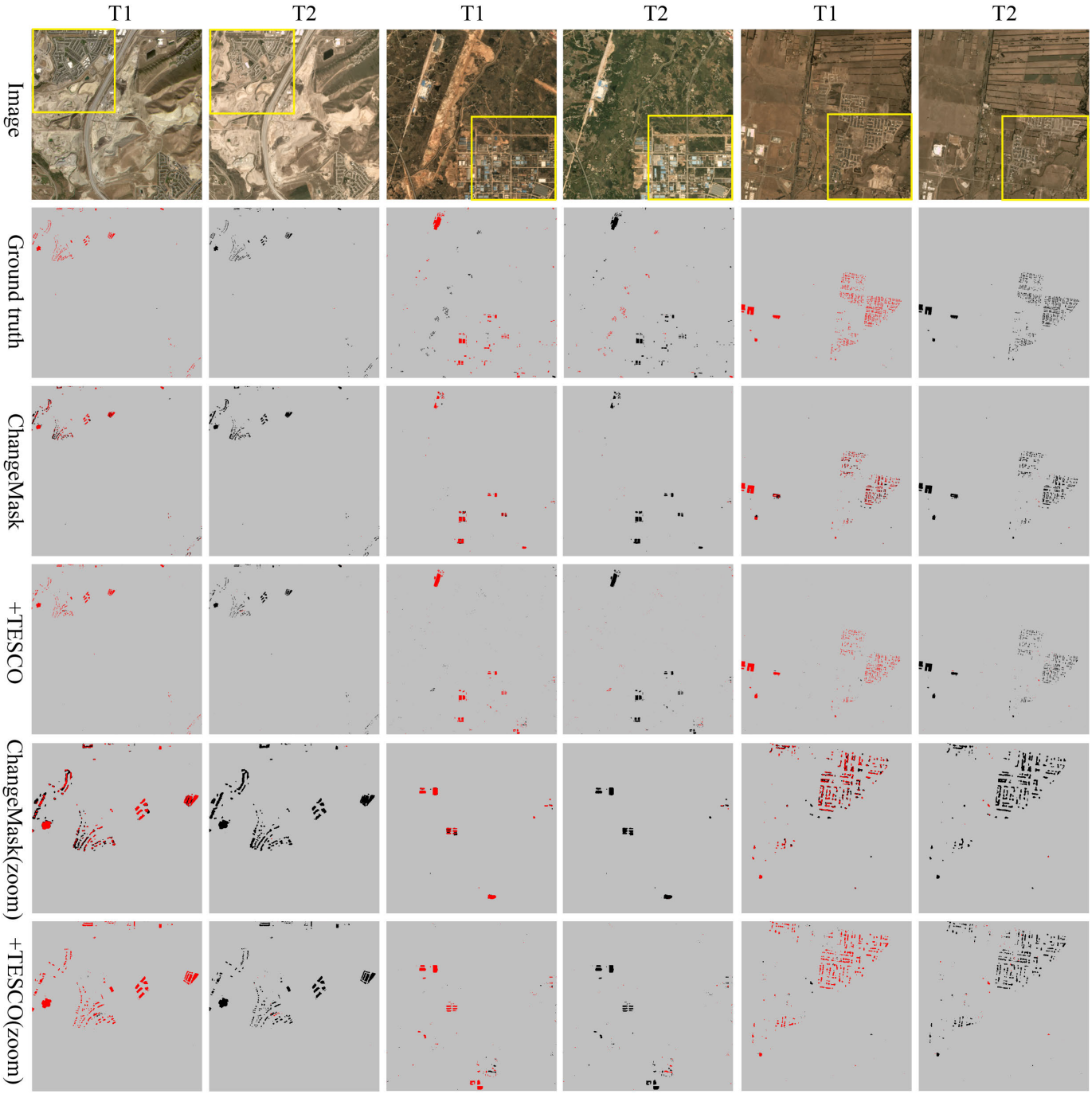


Fig. 8. Visualization of the semantic change results of the ChangeMask with/without the TESCO module in SpaceNet 7 test set. The fourth and fifth rows are the zoomed-in view results of the yellow box in the images. Legend: unchanged, building, and no building.

optimizer on a single NVIDIA Titan RTX GPU with a weight decay of 0.0001 and a momentum of 0.9. The images were cropped into nonoverlapping patches with 512 × 512 pixels for training. The batch size was 4. We used horizontal and vertical flips, and rotation of $90{*}k$ $(k = 1, 2, 3)^\circ$ as training data augmentation.

3) Main Result: As shown in Table III, the accuracy of the semantic segmentation is improved by nearly 1%–2% with the TESCO module. Combining the multitask supervised methods with the TESCO module can improve the performance in both SCD and BCD. The BCD performance is significantly improved for the PCC methods because the TESCO module embeds temporal consistency into the semantic features, predicating identical segmentation results for the unchanged areas. We selected three areas to visualize the semantic segmentation results of ChangeMask with and without the TESCO module in Fig. 7. As shown in the zoomed-in view results for the first area and third area, the TESCO module can improve the completeness of the buildings. The bitemporal results of the TESCO module are more homogeneous in the zoomed-in view results for the second area. Fig. 8 shows the SCD results for these three areas. The zoomed-in view results of the fourth

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TABLE IV
QUANTITATIVE ASSESSMENT RESULTS OF SCD BASELINES WITH/WITHOUT THE TESCO MODULE ON THE DYNAMICEARTHNET DATASET

Method	Backbone	semantic segmentation	Semantic change detection			Binary change detection
		$mIoU_{seg}(\%)$	$mIoU_{sc}(\%)$	$mIoU^{c}_{sc}(\%)$	$mIoU^{nc}_{sc}(\%)$	$IoU_{bc}(\%)$
Multi-task supervised						
ChangeMask [8]	EfficientNet-B0	**36.97**	**20.04**	7.49	32.58	**14.55**
+ TESCO		**38.96**$_{(+1.99)}$	**21.53**$_{(+1.49)}$	8.37	34.68	**17.00**$_{(+2.45)}$
ChangeStar [9]	ResNet-34	**37.18**	**20.37**	7.65	33.09	**16.80**
+ TESCO		**38.51**$_{(+1.33)}$	**21.44**$_{(+1.07)}$	8.40	34.47	**16.94**$_{(+0.14)}$
Segmentation supervised						
U-Net [42]	EfficientNet-B0	**37.32**	**20.47**	7.54	33.41	**16.46**
+ TESCO		**39.15**$_{(+1.83)}$	**21.61**$_{(+1.14)}$	8.51	34.71	**17.21**$_{(+0.75)}$
FarSeg [41]	ResNet-34	**37.29**	**20.33**	7.56	33.10	**16.65**
+ TESCO		**38.58**$_{(+1.29)}$	**21.14**$_{(+0.81)}$	7.70	34.57	**16.58**$_{(-0.07)}$

and fifth lines of Fig. 8 show that after the introduction of the TESCO module, the missed detections in the ChangeMask results are significantly reduced, and the integrity of the SCD results is improved. This indicates that the TESCO module utilizes temporal consistency between temporal imagery to improve semantic segmentation and SCD results.

E. Time-Series SCD

Satellites can acquire rich images with a high temporal resolution to generate sequential change maps for the monitoring of ground objects. The sequential change maps can provide information for predicting change tendency. However, how to obtain convincing sequential change maps with time-series imagery is a challenging task. We conducted experiments on the DynamicEarthNet dataset to evaluate the performance of the TESCO module.

1) DynamicEarthNet Dataset [44]: The DynamicEarthNet dataset is a daily multispectral satellite dataset for SCD. The dataset contains daily satellite data acquired from January 2018 to December 2019 and the pixelwise semantic labels for the first day of each month. The annotations include seven LULC semantic classes: impervious surface, agriculture, forest and other vegetation, wetlands, soil, water, and snow and ice. Unlike the Hi-UCD and SpaceNet 7 datasets, DynamicEarthNet focuses on natural ground objects. Only 7.1% of the labels of the DynamicEarthNet dataset are impervious surface, but 44.9% are forest and other vegetation. Each image has four channels (RGB + near-infrared) with a spatial resolution of 3 m and a pixel size of 1024 × 1024. The DynamicEarthNet dataset covers 75 areas of interest (AOIs) across the globe. There are 55 AOIs in the training set, ten AOIs in the validation set, and ten AOIs in the test set. Due to the labels of the validation and test sets not being released, we randomly selected 44 AOIs in the training set for training, five AOIs for validation, and six AOIs for testing. Since snow and ice does not appear in the test set, we ignored the labels of snow and ice in the training set. Furthermore, we only used the images with semantic labels for each ROI's first day of each month. The change samples for training and testing were generated by comparing the monthly images from January 2018 to December 2019, including 23 change maps for each ROI. Finally, there were 17 semantic changes in the test set.

2) Implementation Details: All the models were trained for 15k iterations with a learning rate policy, where the initial learning rate was set to 0.03 and multiplied by $(1 - (\text{step}/\max \text{step}))^{\gamma}$ with $\gamma = 0.9$. We used SGD as the optimizer on a single NVIDIA GeForce RTX 3090 GPU with a weight decay of 0.0001 and a momentum of 0.9. The batch size was 16. We used horizontal and vertical flips, and rotation of $90{*}k$ $(k = 1, 2, 3)^{\circ}$ as training data augmentation. The images were cropped into nonoverlapping patches with 512 × 512 pixels for training. In order to evaluate the accuracy of the time-series imagery change, we followed the evaluation protocol in [44]. Different methods were first used to predict the monthly segmentation maps from January 2018 to December 2019, to assess the segmentation accuracy. We then compared adjacent months in each ROI to generate change maps to evaluate the accuracy of the SCD and BCD. The segmentation metric $mIoU_{seg}$ is the average of the IoU of 144 segmentation maps in the test set.

3) Main Results: The qualitative assessment results are listed in Table IV. After adding the TESCO module to the baseline methods, most of the metrics increased by nearly 2%. As with the other datasets, this improvement mainly comes from the results for the unchanged area. We visualize the multitemporal semantic segmentation results of U-Net with and without the TESCO module in Fig. 9. Affected by the phenological change of vegetation from May 1, 2019 to October 1, 2019, U-Net cannot accurately classify some areas as agricultural land, while the TESCO module can overcome the phenological change to obtain more complete results. Fig. 10 shows SCD results. Compared with U-Net, the TESCO module can significantly reduce the false alarms in the results while also improving the semantic change category recognition accuracy. Compared with the

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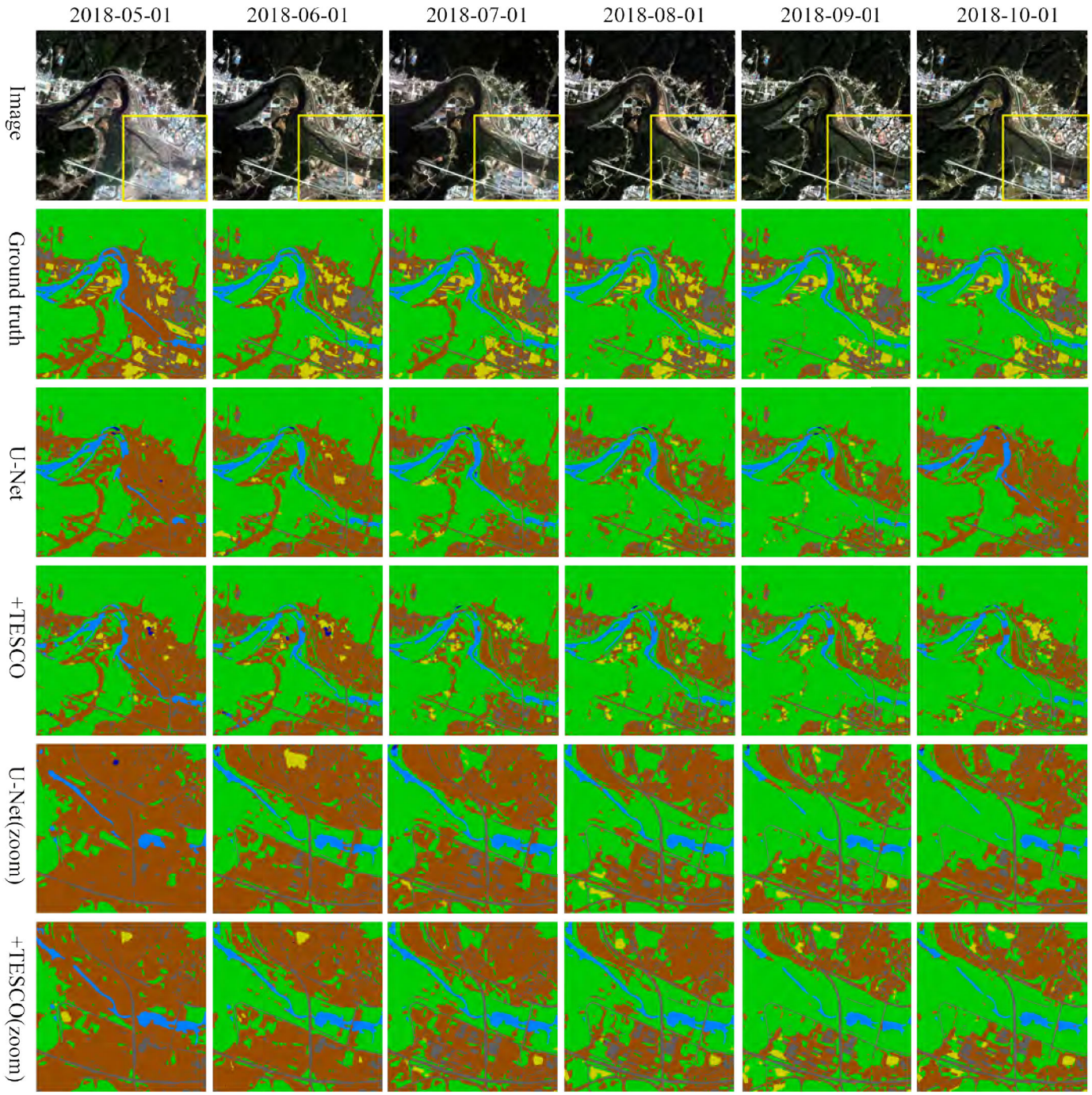


Fig. 9. Visualization of the semantic segmentation results of U-Net with/without the TESCO module on the DynamicEarthNet dataset. The fifth and sixth rows are the zoomed-in view results of the yellow box in images. Legend: impervious surface, agriculture, forest and other vegetation, wetlands, soil, water, and no change.

other datasets, the metric increase on the DynamicEarthNet dataset is slight. The DynamicEarthNet dataset mainly focuses on natural ground objects, and the monthly changes of these ground objects are not obvious in imagery with a meter-level spatial resolution. The ground objects in these areas with similar texture and/or spectral information make it challenging to distinguish change and unchanged areas, resulting in false estimation for the value of W_b. Therefore, the accuracy improvement of the TESCO module becomes less.

V. Discussion and Analysis

A. Ablation Study

In order to explore the role of each part of the TESCO module, we added the TESCO module to FarSeg and ChangeStar to carry out ablation experiments using semantic supervision and multitask supervision.

1) Temporal Fusion: Tables V and VI list the performance of the asymmetric (CatConv) and symmetric temporal fusion operations (temporal difference and TST) in the TESCO module supervised by semantic segmentation and multitask.

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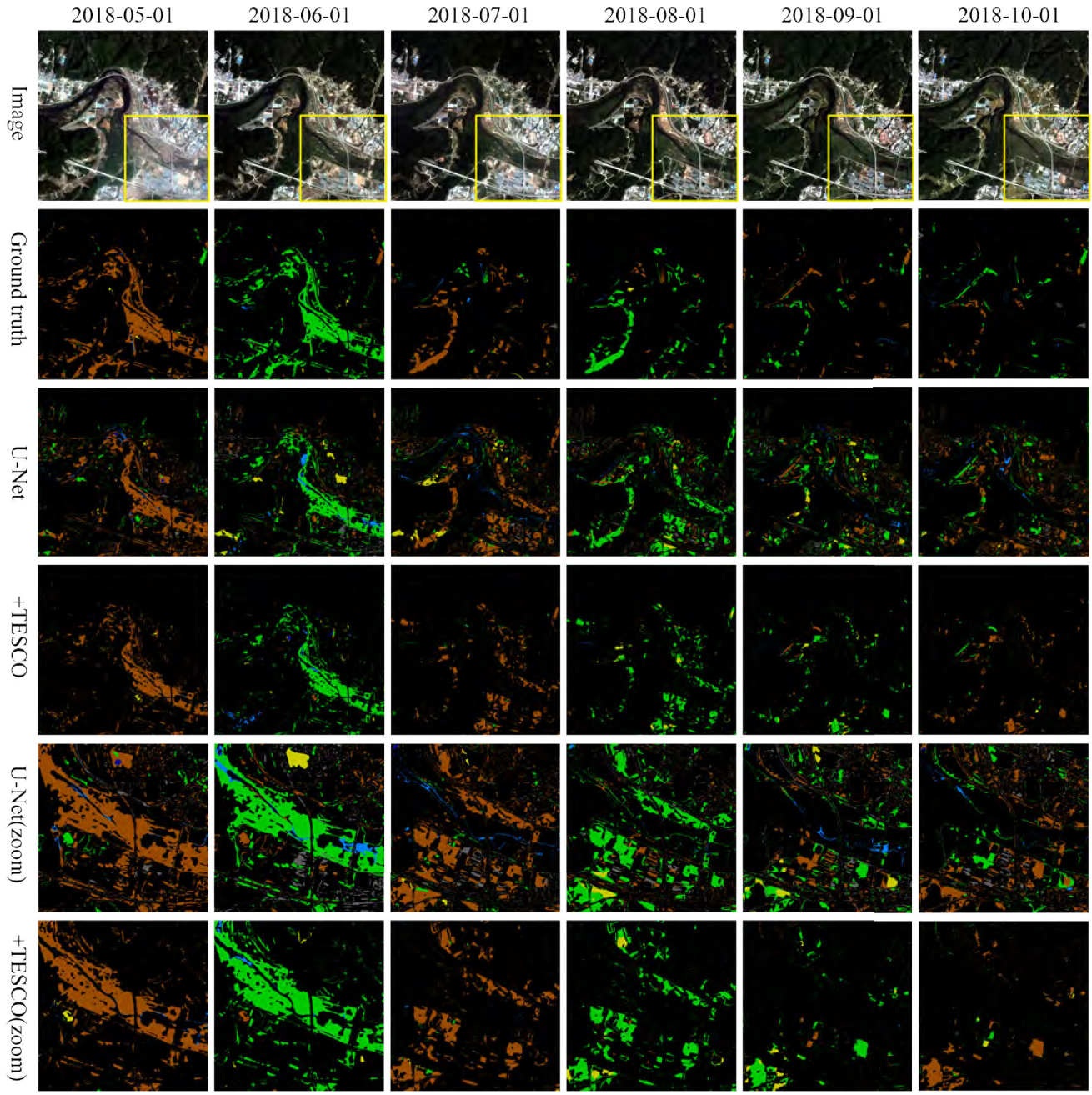


Fig. 10. Visualization of the semantic change results of U-Net with/without the TESCO module on the DynamicEarthNet dataset. The fifth and sixth rows are the zoomed-in view results of the yellow box in images. Legend: impervious surface, agriculture, forest and other vegetation, wetlands, soil, water, and no change.

TABLE V

PERFORMANCE OF DIFFERENT TEMPORAL FUSION OPERATIONS FOR BLENDING WEIGHT ESTIMATION IN THE TESCO MODULE ON THE HI-UCD TEST SET SUPERVISED BY SEMANTIC SEGMENTATION

Methods	Temporal fusion	Semantic segmentation			Semantic change detection			Binary change detection
		$\text{mIoU}_{\text{seg}}(\%)$	$\text{mIoU}_{\text{seg}}^{2018}(\%)$	$\text{mIoU}_{\text{seg}}^{2019}(\%)$	$\text{mIoU}_{\text{sc}}(\%)$	$\text{mIoU}_{\text{sc}}^{\text{c}}(\%)$	$\text{mIoU}_{\text{sc}}^{\text{nc}}(\%)$	$\text{IoU}_{\text{bc}}(\%)$
FarSeg +TESCO	CatConv	**61.43**	61.33	61.52	**35.29**	13.25	57.33	**26.36**
	$\lvert F_{t1} - F_{t2}\rvert$	**61.65**	61.58	61.72	**36.57**	15.84	57.30	**25.15**
	TST ($k = 3$)	**63.09**	62.84	63.33	**36.56**	14.27	58.85	**26.80**

For semantic segmentation supervision, TST achieves the best performance on semantic segmentation and BCD. Temporal difference has the highest score on SCD. For multitask supervision, temporal difference has the highest score on semantic segmentation and SCD while TST has the highest score on BCD. Generally, symmetric fusion performs better

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TABLE VI

PERFORMANCE OF DIFFERENT TEMPORAL FUSION OPERATIONS FOR BLENDING WEIGHT ESTIMATION IN THE TESCO MODULE ON THE HI-UCD TEST SET SUPERVISED BY MULTITASK

Methods	Temporal fusion	Semantic segmentation			Generation of change	Semantic change detection			Binary change detection	Semantic change consistency
		$mIoU_{seg}(\%)$	$mIoU^{2018}_{seg}(\%)$	$mIoU^{2019}_{seg}(\%)$		$mIoU_{sc}(\%)$	$mIoU^{c}_{sc}(\%)$	$mIoU^{nc}_{sc}(\%)$	$IoU_{bc}(\%)$	$mIoU_{scc}$
ChangeStar + TESCO	CatConv	**57.60**	57.57	57.62	MASK	**33.41**	12.99	53.84	**26.83**	**77.31**
					PCC	**34.28**	14.53	54.04	**28.84**	
	$\|F_{t1} - F_{t2}\|$	**59.66**	59.36	59.97	MASK	**34.62**	13.52	55.71	**27.08**	**74.72**
					PCC	**35.66**	15.34	55.97	**29.76**	
	TST ($k = 3$)	**56.65**	56.73	56.56	MASK	**33.72**	14.04	53.39	**31.24**	**77.13**
					PCC	**34.85**	16.27	53.43	**31.68**	

TABLE VII

INFLUENCE OF THE BLENDING DIRECTION AND THE VALUE RANGE OF W_b FOR THE TSFB ON THE HI-UCD TEST SET SUPERVISED BY SEMANTIC SEGMENTATION

Methods	Blending direction	Range of W_b	Semantic segmentation			Semantic change detection			Binary change detection
			$mIoU_{seg}(\%)$	$mIoU^{2018}_{seg}(\%)$	$mIoU^{2019}_{seg}(\%)$	$mIoU_{sc}(\%)$	$mIoU^{c}_{sc}(\%)$	$mIoU^{nc}_{sc}(\%)$	$IoU_{bc}(\%)$
FarSeg +TESCO	T1 $\rightarrow$ T2	[0,1]	**58.12**	57.34	58.90	**32.92**	12.06	53.77	**20.91**
	T2 $\rightarrow$ T1	[0,1]	**57.27**	56.18	58.37	**32.45**	12.08	52.82	**20.26**
	T1 $\leftrightarrow$ T2	[0,1]	**62.28**	61.99	62.57	**35.46**	13.41	57.51	**23.23**
	T1 $\leftrightarrow$ T2	[0.5,1]	**61.16**	60.96	61.37	**36.44**	15.81	57.07	**26.38**

TABLE VIII

INFLUENCE OF THE BLENDING DIRECTION AND THE VALUE RANGE OF W_b FOR THE TSFB ON THE HI-UCD TEST SET SUPERVISED BY MULTITASK

Methods	Blending direction / Range of W_b	Semantic segmentation			Generation of change	Semantic change detection			Binary change detection	Semantic change consistency
		$mIoU_{seg}(\%)$	$mIoU^{2018}_{seg}(\%)$	$mIoU^{2019}_{seg}(\%)$		$mIoU_{sc}(\%)$	$mIoU^{c}_{sc}(\%)$	$mIoU^{nc}_{sc}(\%)$	$IoU_{bc}(\%)$	$mIoU_{scc}$
ChangeStar + TESCO	T1 $\rightarrow$ T2 [0,1]	**58.02**	58.13	57.92	MASK	**33.99**	13.29	54.69	**30.75**	**69.42**
					PCC	**32.95**	11.70	54.19	**25.79**	
	T2 $\rightarrow$ T1 [0,1]	**53.42**	52.89	53.95	MASK	**31.56**	12.95	50.17	**28.13**	**57.67**
					PCC	**29.97**	10.74	49.20	**18.48**	
	T1 $\leftrightarrow$ T2 [0,1]	**56.73**	55.93	57.53	MASK	**32.67**	12.76	52.58	**28.03**	**61.03**
					PCC	**32.29**	12.86	51.73	**19.55**	
	T1 $\leftrightarrow$ T2 [0.5,1]	**57.60**	57.57	57.62	MASK	**33.41**	12.99	53.84	**26.83**	**77.31**
					PCC	**34.28**	14.53	54.04	**28.84**	

TABLE IX

ACCURACY OF DIFFERENT ξ FOR THE SPATIOTEMPORAL FEATURE UPDATE ON THE HI-UCD TEST SET SUPERVISED BY SEMANTIC SEGMENTATION

Model	Loops	ST Feature update	Semantic segmentation			Semantic change ddetection			Binary change detection
			$mIoU_{seg}$	$mIoU^{2018}_{seg}(\%)$	$mIoU^{2019}_{seg}(\%)$	$mIoU_{sc}$	$mIoU^{c}_{sc}(\%)$	$mIoU^{nc}_{sc}(\%)$	$IoU_{bc}(\%)$
FarSeg	-	-	**60.36**	60.00	60.72	**31.43**	9.90	52.96	**8.54**
+ TESCO	R=1	-	**61.16**	60.96	61.37	**36.44**	15.81	57.07	**26.38**
	R=2	$\xi = F'_t$	**61.84**	61.44	62.25	**36.59**	15.54	57.63	**26.34**
		$\xi = F_t + F'_t$	**62.23**	62.10	62.37	**37.46**	16.96	57.97	**27.68**
		$\xi = (F_t + F'_t)/2$	**62.31**	61.97	62.65	**36.53**	14.93	58.13	**26.71**
		ξ =CatConv(F'_t,F_t)	**62.63**	62.63	63.41	**36.56**	14.28	58.84	**27.29**

because it can generate more distinguishable spatiotemporal features, which is beneficial for estimating the appropriate value of blending weight.

2) Temporal-Semantic Feature Blending: In this section, we discuss the effect of the blending direction and the influence of the value range of W_b. As shown in

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TABLE X

ACCURACY OF DIFFERENT ξ FOR THE SPATIOTEMPORAL FEATURE UPDATE ON THE HI-UCD TEST SET SUPERVISED BY MULTITASK

Model	Loops	ST Feature update	Semantic segmentation			Generation of change	Semantic change detection			Binary change detection	Semantic change consistency
			$mIoU_{seg}$	$mIoU_{seg}^{2018}(\%)$	$mIoU_{seg}^{2019}(\%)$		$mIoU_{sc}$	$mIoU_{sc}^{c}(\%)$	$mIoU_{sc}^{nc}(\%)$	$IoU_{bc}(\%)$	$mIoU_{scc}(\%)$
ChangeStar	-	-	**56.31**	55.56	57.05	MASK	**31.92**	12.73	51.12	**27.67**	**46.63**
						PCC	**29.13**	8.99	49.27	**9.23**	
+ TESCO	R=1	-	**57.60**	57.57	57.62	MASK	**33.41**	12.99	53.84	**26.83**	**77.31**
						PCC	**34.28**	14.53	54.04	**28.84**	
	R=2	$\xi = F_t'$	**58.06**	57.84	58.28	MASK	**33.81**	13.31	54.31	**27.32**	**78.42**
						PCC	**34.44**	14.42	54.46	**28.91**	
		$\xi = F_t + F_t'$	**58.72**	58.37	59.08	MASK	**33.68**	12.55	54.81	**27.97**	**80.75**
						PCC	**35.05**	15.02	55.07	**30.60**	
		$\xi = (F_t + F_t')/2$	**57.53**	57.58	57.49	MASK	**33.36**	13.00	53.73	**26.84**	**77.74**
						PCC	**34.28**	14.48	54.07	**29.50**	
		ξ =CatConv(F_t',F_t)	**58.28**	58.35	58.22	MASK	**33.57**	12.40	54.75	**30.68**	**79.58**
						PCC	**33.51**	12.37	54.65	**29.68**	

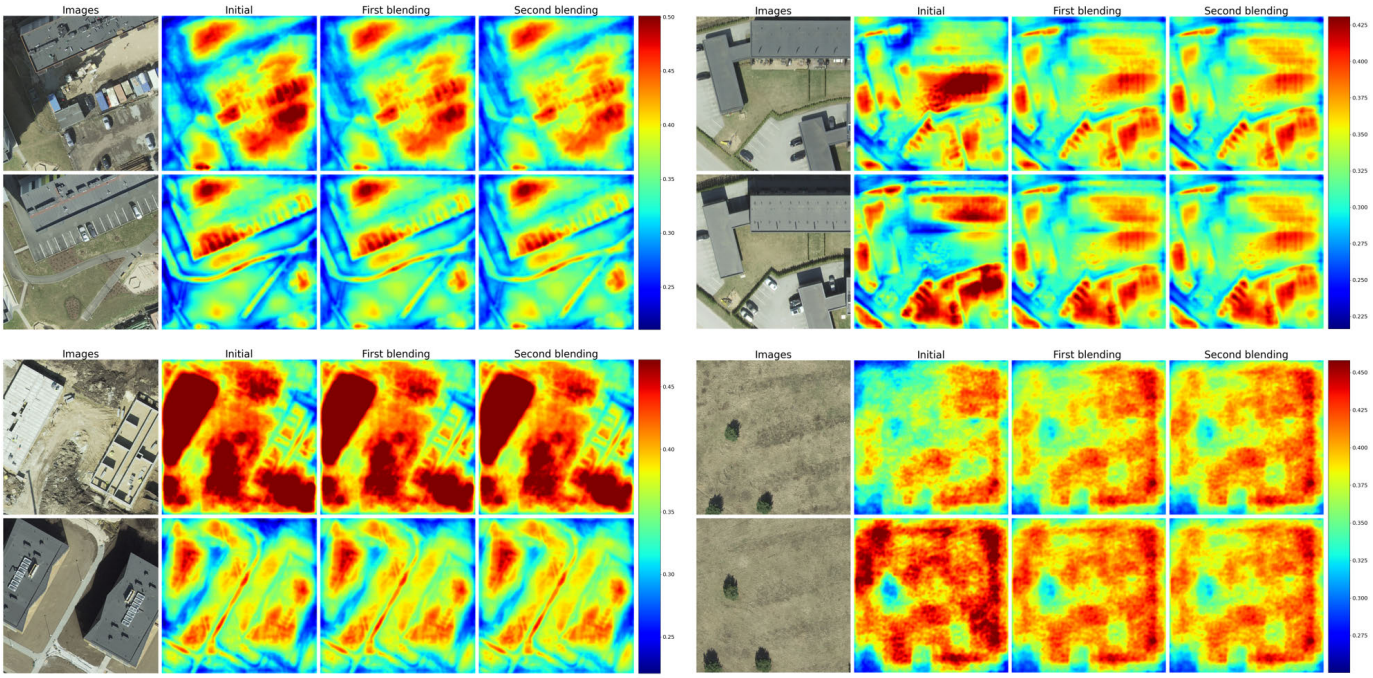


Fig. 11. Visualization of the semantic features before and after the recurrence of TESCO module [R = 2 and $\xi = (F_t + F_t')$] on FarSeg. The features of the first row are in T1, and the features of the second row are in T2. (a) Changed area. (b) Unchanged area.

Tables VII and VIII, bidirectional blending performs better in both segmentation and change detection because the temporal consistency is temporally symmetric. Meanwhile, bidirectional blending can increase the interaction between bitemporal semantic features. As analyzed in Section III-B1, the optimal value of W_b for the unchanged area is 0.5 when using bidirectional blending. After using ReLU to change the value range of W_b from [0, 1] to [0.5, 1], the BCD performance increases by 3.15% and the SCD performance increases by nearly 1% in Table VII. While using multitask supervision, the semantic change consistency achieves the highest score in Table VIII, and the result generated by PCC is better than MASK. Therefore, an appropriate value of W_b is the key to maintaining temporal consistency and improving the SCD results.

3) Spatiotemporal Feature Update: Tables IX and X list the accuracy when we cycle the TESCO module with different ξ for PCC and multitask methods. The performance of the three tasks is improved when recycling the TESCO module because it optimizes the temporal correlation of semantic features from coarse to fine. When $\xi = F_t + F_t'$, $mIoU_{sc}$ is increased by nearly 1% in Table IX, achieving the best results after adding TESCO module to FarSeg. Similarly, incorporating TESCO module to ChangeStar, the semantic change consistency is improved by 34.12%. Meanwhile, semantic segmentation and SCD also achieved the highest scores. As shown in Figs. 11 and 12, the feature response for the unchanged areas gradually becomes more similar with the recurrence of the TESCO module. In contrast, the features of the changed regions remain discriminative. Therefore, the TESCO module can help the

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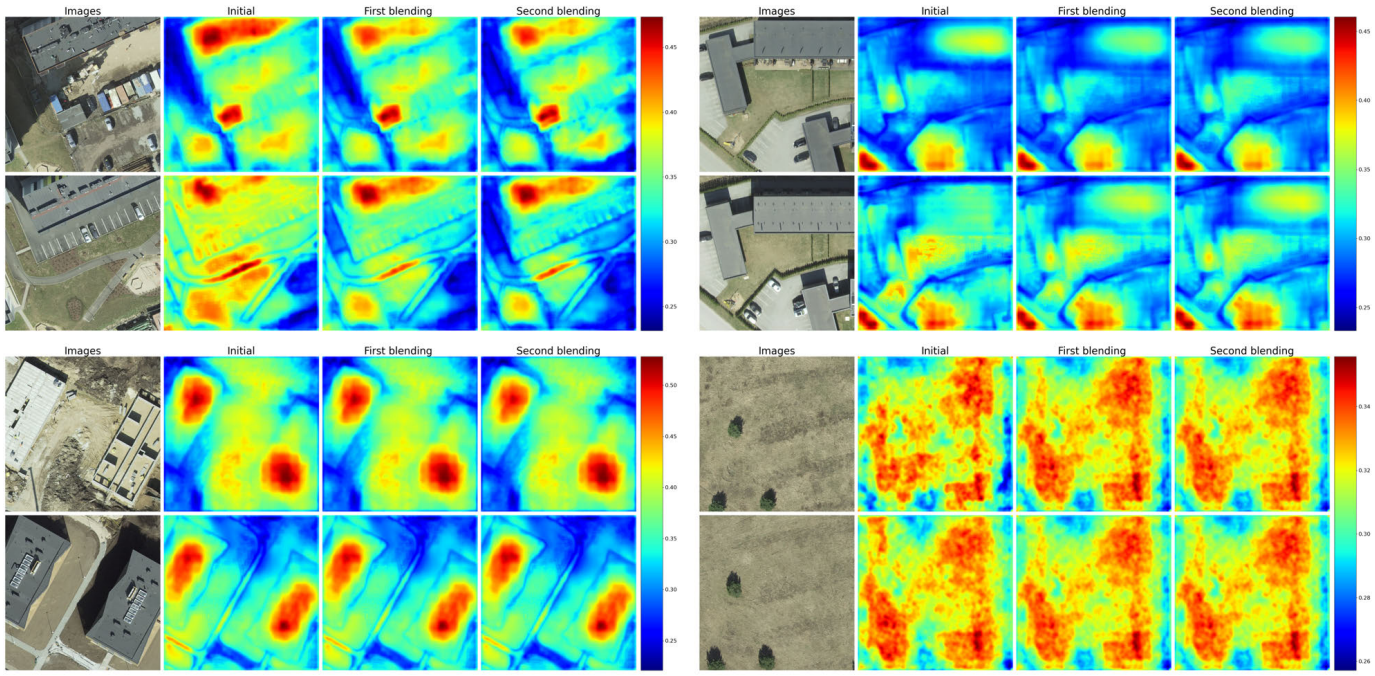


Fig. 12. Visualization of the semantic features before and after the recurrence of TESCO module [$R = 2$ and $\xi = (F_t + F_t')$] on ChangeStar. The features of the first row are in T1, and the features of the second row are in T2. (a) Changed area. (b) Unchanged area.



Fig. 13. Optimal loop value (R) for the TESCO module in multitask and PCC-based SCD methods. (a) Accuracy of ChangeStar with TESCO. (b) Accuracy of FarSeg with TESCO.

network to learn temporal consistency from coarse to fine and achieve better SCD performance.

4) Optimal Number of Loops for the TESCO Module: With the aim of identifying the optimal value of the loop parameter (R), TESCO modules with R values ranging from 1 to 10 were subjected to testing. The accuracies of the ChangeStar and FarSeg when integrated with the TESCO module are illustrated in Fig. 13. For the ChangeStar method [Fig. 13(a)], repeated cycling of the TESCO module leads to an improvement in semantic change consistency (mIoU_{scc}). However, as R increases, a decline in the accuracy of semantic segmentation (mIoU_{seg}) and SCD (mIoU_{sc}) is observable. For the FarSeg method [Fig. 13(b)], the accuracy of each task fluctuates as R increases. The accuracy of each task exhibits fluctuations as R increases. Irrespective of the supervised model employed, a more favorable tradeoff among the three tasks of semantic segmentation, BCD, and SCD is generally achieved when $R = 2$. Therefore, it is recommended that R should be set to 2 when cycling the TESCO module.

B. Comparing the TESCO Module With Other Methods

To verify the advancement of the proposed module, we added other comparative methods considering temporal correlation to ChangeStar and FarSeg, i.e., CG [13], CA [14], Cot-SR [15], CSLoss [15], SCanFormer [37], PsdLoss [37], SFE [36], and TCRPN [10].

Table XI lists the evaluation results. For the multitask methods, the existing methods cannot simultaneously improve the performance of segmentation, SCD, and BCD. However, after adding the TESCO module, the metrics for all three tasks show a 1%–2% improvement when $R = 1$. Furthermore, when $R = 2$, the scores of the three tasks are further improved. For

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TABLE XI
COMPARING THE ACCURACY OF THE TESCO MODULE WITH OTHER TEMPORAL CONSISTENCY METHODS ON THE HI-UCD TEST SET

Method	Semantic segmentation			Semantic change detection			Binary change detection	Params (M)	MAdds (B)
	$mIoU_{seg}$(%)	$mIoU_{seg}^{2018}$(%)	$mIoU_{seg}^{2019}$(%)	$mIoU_{sc}$(%)	$mIoU_{sc}^{c}$(%)	$mIoU_{sc}^{nc}$(%)	IoU_{bc}(%)		
Multi-task supervised									
ChangeStar [9]	**56.31**	55.56	57.05	**31.92**	12.73	51.12	**27.67**	26.51	21.18
+ CG [13]	**58.25**$_{(+1.94)}$	58.02	58.48	**31.86**$_{(-0.06)}$	9.44	54.28	**28.08**$_{(+0.41)}$	26.51$_{(+0.00)}$	21.18$_{(+0.00)}$
+ CA [14]	**57.92**$_{(+1.61)}$	57.83	58.00	**31.92**$_{(+0.00)}$	9.84	54.00	**27.47**$_{(-0.20)}$	26.51$_{(+0.00)}$	21.20$_{(+0.02)}$
+ Cot-SR [15]	**60.93**$_{(+4.62)}$	60.60	61.26	**34.17**$_{(+2.25)}$	11.59	56.75	**23.87**$_{(-3.80)}$	26.54$_{(+0.03)}$	25.04$_{(+3.84)}$
+ CSLoss [15]	**53.74**$_{(-2.57)}$	51.88	55.59	**31.71**$_{(-0.21)}$	13.61	49.81	**27.32**$_{(-0.35)}$	26.51$_{(+0.00)}$	21.18$_{(+0.00)}$
+ SCanFormer [37]	**56.41**$_{(+0.10)}$	55.94	56.87	**31.71** $_{(-0.21)}$	10.51	52.91	**33.43** $_{(+5.76)}$	28.60$_{(+2.09)}$	30.01$_{(+8.83)}$
+ PsdLoss [37]	**56.72**$_{(+0.41)}$	55.79	57.65	**33.55** $_{(+1.63)}$	15.00	52.10	**27.74** $_{(+0.08)}$	26.51$_{(+0.00)}$	21.18$_{(+0.00)}$
+ SFE [36]	**55.65**$_{(+0.10)}$	54.34	56.96	**32.20**$_{(-0.21)}$	13.75	50.65	**29.42**$_{(+5.76)}$	26.51$_{(+0.00)}$	21.18$_{(+0.00)}$
+ TCRPN [10]	**55.50**$_{(-0.81)}$	55.01	55.99	**33.32**$_{(+1.40)}$	14.53	52.10	**29.05**$_{(+1.38)}$	26.51$_{(+0.00)}$	21.18$_{(+0.00)}$
+ TESCO (R=1)	**57.60**$_{(+1.29)}$	57.57	57.62	**34.28**$_{(+2.36)}$	14.53	54.04	**28.84**$_{(+1.17)}$	26.51$_{(+0.00)}$	21.87$_{(+0.69)}$
+ TESCO (R=2)	**58.72**$_{(+2.14)}$	58.37	59.08	**35.05**$_{(+3.13)}$	15.02	55.07	**30.60**$_{(+2.93)}$	26.51$_{(+0.00)}$	22.55$_{(+1.37)}$
Semantic supervised									
FarSeg [41]	**60.36**	60.00	60.72	**31.43**	9.90	52.96	**8.54**	26.34	20.50
+ CG [13]	**60.90**$_{(+0.54)}$	60.32	61.49	**32.44**$_{(+1.01)}$	9.23	55.65	**16.63**$_{(+8.09)}$	26.51$_{(+0.17)}$	21.18$_{(+0.68)}$
+ CA [14]	**60.92**$_{(+0.56)}$	60.50	61.34	**32.81**$_{(+1.38)}$	9.66	55.96	**17.71**$_{(+9.17)}$	26.51$_{(+0.17)}$	21.19$_{(+0.69)}$
+ Cot-SR [15]	**62.50**$_{(+2.14)}$	62.14	62.86	**33.06**$_{(+1.63)}$	9.43	56.70	**10.57**$_{(+2.03)}$	26.37$_{(+0.03)}$	24.36$_{(+3.86)}$
+ CSLoss [15]	**61.03**$_{(+0.67)}$	60.83	61.24	**31.46**$_{(+0.03)}$	9.56	53.37	**8.80**$_{(+0.26)}$	26.34$_{(+0.00)}$	20.50$_{(+0.00)}$
+ SCanFormer [37]	**61.10**$_{(+0.74)}$	60.54	61.66	**33.34**$_{(+1.91)}$	9.85	56.83	**20.61**$_{(+12.07)}$	28.60$_{(+2.26)}$	30.01$_{(+9.51)}$
+ PsdLoss [37]	**61.36**$_{(+1.00)}$	60.24	62.48	**32.13**$_{(+0.70)}$	10.43	53.83	**8.50**$_{(-0.04)}$	26.51$_{(+0.00)}$	20.50$_{(+0.00)}$
+ SFE [36]	**61.78**$_{(+1.42)}$	61.42	62.15	**31.91**$_{(+0.48)}$	10.02	53.81	**8.47**$_{(-0.07)}$	26.51$_{(+0.17)}$	21.18$_{(+0.68)}$
+ TCRPN [10]	**61.03**$_{(+0.67)}$	60.83	61.24	**31.46**$_{(+0.03)}$	9.56	53.37	**8.80**$_{(+0.26)}$	26.51$_{(+0.17)}$	21.18$_{(+0.68)}$
+ TESCO (R=1)	**61.16**$_{(+0.80)}$	60.96	61.37	**36.44**$_{(+5.01)}$	15.81	57.07	**26.38**$_{(+17.84)}$	26.51$_{(+0.21)}$	21.19$_{(+0.69)}$
+ TESCO (R=2)	**62.23**$_{(+1.87)}$	62.10	62.37	**37.46**$_{(+6.03)}$	16.96	57.97	**27.68**$_{(+19.14)}$	26.51$_{(+0.21)}$	21.87$_{(+1.37)}$

the PCC methods, the substantial improvement of $mIoU_{sc}^{nc}$ in all the methods improves the detection effect for the SCD and BCD. This phenomenon suggests that learning temporal consistency improves the performance of segmentation networks when dealing with multitemporal images. However, only the TESCO module can significantly improve the $mIoU_{sc}^{c}$ score in the PCC methods, achieving the highest $mIoU_{sc}$ and IoU_{bc} scores. This demonstrates that explicitly modeling temporal consistency by the TESCO module is superior to other methods and is more robust to radiation differences and geometric errors in different images. At the same time, after adding the TESCO module, FarSeg's semantic segmentation and semantic change accuracy exceeded the ChangeStar method. Comparing the other methods with CSLoss and PsdLoss shows that considering temporal consistency at the feature level works better than considering it at the loss level.

C. Maintaining Semantic Change Consistency

We compared the TESCO module with other methods to evaluate the ability to maintain semantic change consistency in multitask methods. The qualitative results on the Hi-UCD test dataset are listed in Table XII. Although most of the methods can improve the consistency of the semantic changes, the TESCO module obtains the highest $mIoU_{scc}$ score, which increases by nearly 31% when recurrence is 1. The $mIoU_{scc}^{c}$ score increases by nearly 50%. After cycling the TESCO module, the score of $mIoU_{scc}$ is further improved. As shown in Fig. 14, the TESCO module effectively suppresses the false alarms in the PCC results, leading to an increase in the $mIoU_{sc}^{nc}$ score by nearly 5% on the Hi-UCD dataset. Furthermore, the TESCO module is the only method with better PCC results than MASK for the multitask SCD methods. Therefore, we can use PCC to generate the final result to maintain semantic change consistency.

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TABLE XII

SEMANTIC CHANGE CONSISTENCY FOR DIFFERENT TEMPORAL CONSISTENCY MODULES IN CHANGESTAR ON THE HI-UCD TEST SET. THE CHANGE RESULTS GENERATED BY THE BINARY CHANGE MASK ARE DENOTED AS "MASK," AND "PCC" REPRESENTS THE CHANGE RESULTS GENERATED BY COMPARING THE BITEMPORAL SEGMENTATION RESULTS

Method	Change generate		Semantic change detection			Binary change detection	Semantic change consistency		
	MASK	PCC	$mIoU_{sc}$(%)	$mIoU^{c}_{sc}$(%)	$mIoU^{nc}_{sc}$(%)	IoU_{bc}(%)	$mIoU_{scc}$(%)	IoU^{c}_{scc}(%)	IoU^{nc}_{scc}(%)
ChangeStar [9]	✓		**31.92**	12.73	51.12	**27.67**	**46.63**	5.87	87.40
		✓	**29.13**	8.99	49.27	**9.23**			
+ CG [13]	✓		**31.86**	9.44	54.28	**28.08**	**57.71**$_{(+11.08)}$	18.45	96.97
		✓	**31.04**	9.06	53.03	**15.68**			
+ CA [14]	✓		**31.92**	9.84	54.00	**27.47**	**60.44**$_{(+13.81)}$	23.07	97.80
		✓	**31.08**	9.07	53.09	**18.55**			
+ Cot-SR [15]	✓		**34.17**	11.59	56.75	**23.87**	**49.62**$_{(+2.99)}$	7.02	92.23
		✓	**31.85**	8.22	55.47	**10.55**			
+ CSLoss [15]	✓		**31.71**	13.61	49.81	**27.32**	**46.77**$_{(+0.14)}$	6.06	87.48
		✓	**28.30**	8.60	48.00	**9.28**			
+ SCanFormer [37]	✓		**31.71**	10.51	52.91	**33.43**	**68.36** $_{(+21.73)}$	38.37	98.36
		✓	**30.22**	8.53	51.91	**23.48**			
+PsdLoss [37]	✓		**33.55**	15.00	52.10	**27.74**	**46.91**$_{(+0.28)}$	6.13	87.69
		✓	**29.20**	8.18	50.23	**9.11**			
+ SFE [36]	✓		**32.20**	13.75	50.65	**29.42**	**46.53**$_{(-0.10)}$	6.16	86.90
		✓	**28.89**	9.17	48.62	**9.01**			
+ TCRPN [10]	✓		**33.32**	14.53	52.10	**29.05**	**47.62** $_{(+0.99)}$	6.86	88.38
		✓	**29.64**	9.12	50.16	**9.63**			
+ TESCO (R=1)	✓		**33.41**	12.99	53.84	**26.83**	**77.31**$_{(+30.68)}$	55.30	99.33
		✓	**34.28**	14.53	54.04	**28.84**			
+ TESCO (R=2)	✓		**33.68**	12.55	54.81	**27.97**	**80.75**$_{(+34.12)}$	61.99	99.52
		✓	**35.05**	15.02	55.07	**30.60**			

TABLE XIII

QUALITATIVE ASSESSMENT RESULTS OF SCD BASELINES WITH/WITHOUT THE TESCO MODULE ON THE SECOND DATASET

Method	Backbone	mIoU (%)	IoU_{bc} (%)	SeK(%)	Score(%)
Multi-task supervised					
ChangeMask [8]	EfficientNet-B0	72.79	57.98	21.02	36.55
+ TESCO		72.48$_{(-0.31)}$	57.47$_{(-0.51)}$	20.74$_{(-0.28)}$	36.26$_{(-0.29)}$
ChangeStar [9]	ResNet-34	70.75	54.72	18.47	34.16
+ TESCO		71.00$_{(+0.25)}$	55.13$_{(+0.41)}$	18.79$_{(+0.32)}$	34.45$_{(+0.29)}$
Segmentation supervised					
U-Net [42]	EfficientNet-B0	50.19	35.72	9.23	21.52
+ TESCO		50.21$_{(+0.02)}$	35.76$_{(+0.04)}$	9.11$_{(-0.12)}$	21.44$_{(-0.08)}$
FarSeg [41]	ResNet-34	49.14	34.59	7.76	20.17
+ TESCO		49.36$_{(+0.22)}$	34.75$_{(+0.16)}$	7.93$_{(+0.17)}$	20.36$_{(+0.19)}$

D. Influence of the Semantic Label in Unchanged Area

The TESCO module mainly learns the consistency of unchanged regions through semantic supervision, thereby overcoming the effects of radiometric and geometric differences in multitemporal images and enhancing the accuracy of SCD. The impact of not providing semantic supervision for

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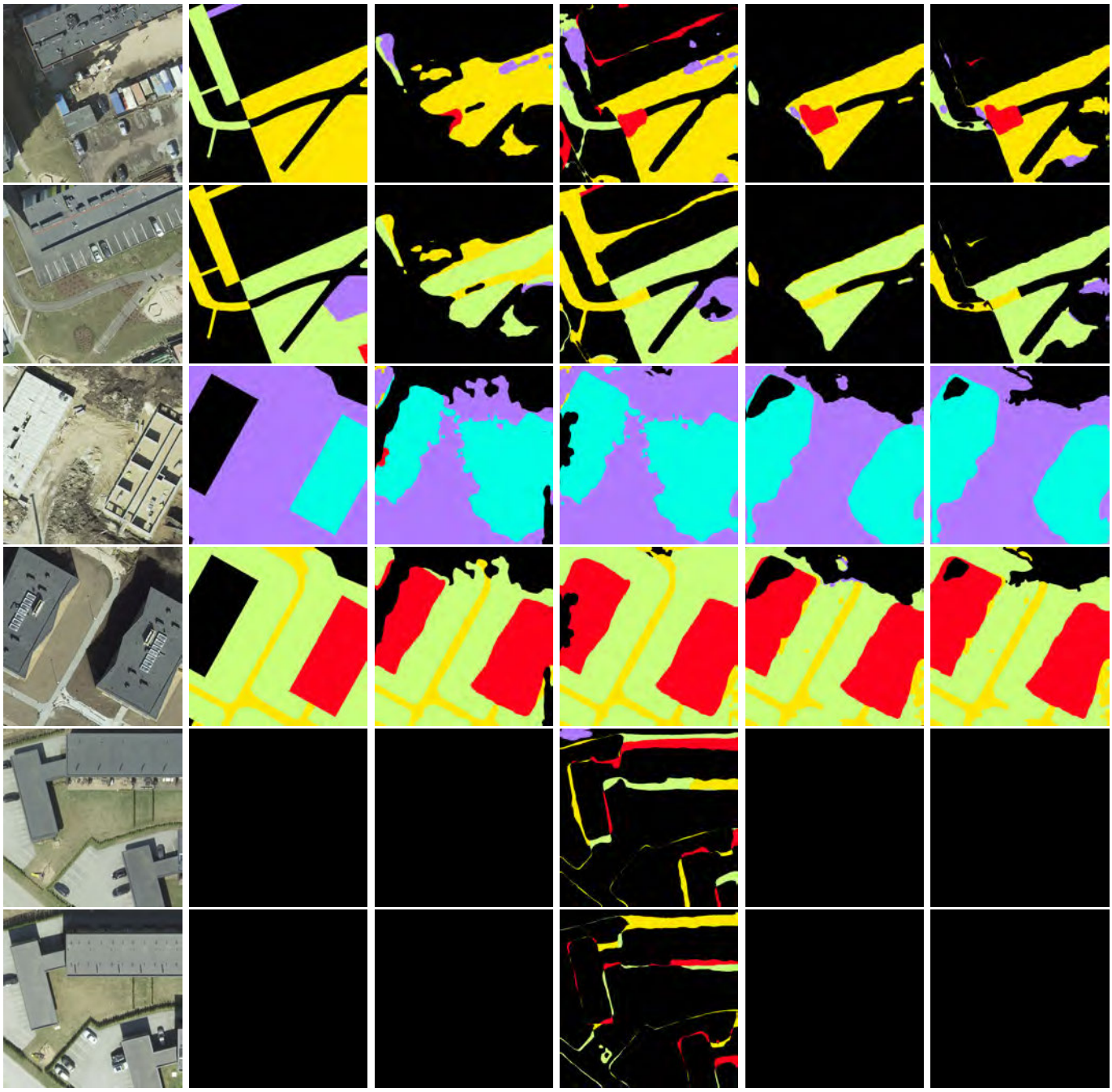


Fig. 14. Visualization of the SCD results generated by PCC and MASK with/without the TESCO module ($R = 1$) in ChangeStar. There is a noticeable performance gap between PCC and MASK on the baseline, resulting in semantic change inconsistency. However, the TESCO module significantly improves the performance of PCC to maintain semantic change consistency. Legend: water, grass, building, greenhouse, road, bridge, bare land, woodland, other, unchanged, and white is unlabeled. (a) Image. (b) Ground truth. (c) ChangeStar (MASK). (d) ChangeStar (PCC). (e) + TESCO (MASK). (f) + TESCO (PCC).

unchanged regions on models remains unknown. To answer this question, we conduct experiments on the SECOND dataset [11].

1) SECOND Dataset [11]: The SECOND dataset is a multicity SCD dataset, which only has the semantic labels in changed areas. To ensure data diversity, it includes 4662 pairs of aerial images. The size of each image is 512 × 512 pixels. The SECOND dataset focuses on six land-cover classes that are frequently found in natural and man-made geographical changes, i.e., nonvegetated ground surface, tree, low vegetation, water, buildings, and playgrounds. There are 30 common change categories (including nonchange) in the SECOND dataset. We followed the official dataset division in [11] to carry out the experiments.

2) Implementation Details: All the models were trained for 15k iterations with a poly learning rate policy, where the initial learning rate was set to 0.01 and multiplied by $(1 - (\text{step}/\max \text{step}))^{\gamma}$ with $\gamma = 0.9$. We used SGD as

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TABLE XIV
QUALITATIVE ASSESSMENT RESULTS ON THE HI-UCD TEST SET OF THE PCC METHOD WITH/WITHOUT THE TESCO MODULE, TRAINED BY DIFFERENT SAMPLE RATIOS

Method	Semantic segmentation			Semantic change detection			Binary change detection
	$mIoU_{seg}(\%)$	$mIoU_{seg}^{2018}(\%)$	$mIoU_{seg}^{2019}(\%)$	$mIoU_{sc}(\%)$	$mIoU_{sc}^{c}(\%)$	$mIoU_{sc}^{nc}(\%)$	$mIoU_{bc}(\%)$
Only changed samples							
FarSeg [41]	**27.38**	29.11	25.65	**12.44**	3.17	21.70	**4.12**
+ TESCO	**26.36**	27.50	25.22	**12.65**	3.12	22.18	**4.30**
Changed: Unchanged = 1:0.5							
FarSeg [41]	**46.56**	43.77	49.35	**23.39**	7.18	39.61	**6.65**
+ TESCO	**53.38**	53.23	53.53	**29.29**	9.50	49.07	**21.08**
Changed: Unchanged = 1:1							
FarSeg [41]	**48.87**	45.08	50.97	**24.54**	7.10	41.98	**7.08**
+ TESCO	**53.20**	52.74	53.66	**29.68**	10.24	49.13	**21.08**
Changed: Unchanged = 1:2							
FarSeg [41]	**49.98**	47.35	52.61	**25.23**	7.36	43.09	**6.99**
+ TESCO	**53.03**	53.26	52.81	**29.64**	10.93	48.36	**20.62**
Changed: Unchanged = 1:4							
FarSeg [41]	**51.19**	47.35	52.61	**26.27**	8.65	43.90	**7.18**
+ TESCO	**53.06**	53.26	52.81	**29.92**	10.98	48.86	**21.70**
Changed: Unchanged ≈ 1:9							
FarSeg [41]	**52.33**	50.91	53.75	**26.79**	8.43	45.15	**7.53**
+ TESCO	**53.58**	53.30	53.84	**30.86**	12.36	49.36	**21.01**
Changed: Unchanged ≈ 1:70							
FarSeg [41]	**60.36**	60.00	60.72	**31.43**	9.90	52.96	**8.54**
+ TESCO	**61.16**	60.96	61.37	**36.44**	15.81	57.07	**26.38**

the optimizer on a single Titan RTX GPU with a weight decay of 0.0001 and a momentum of 0.9. The batch size was 16. We used horizontal and vertical flips, and rotation of $90{*}k$ $(k = 1, 2, 3)^\circ$ as training data augmentation. The results of all multitask supervised methods are generated by MASK.

3) Metrics: In order to follow the official metrics, we adopted the separated kappa (SeK) coefficient proposed in [11] for the SECOND dataset. To comprehensively consider the binary and semantic change, a weighted overall score is adopted, which is defined as follows:

$$\text{mIoU} = (\text{IoU}_{\text{bc}} + \text{IoU}_{\text{nc}})/2 \tag{13}$$

$$\text{Score} = 0.3 \cdot \text{mIoU} + 0.7 \cdot \text{SeK}. \tag{14}$$

4) Main Results: Table XIII lists the SCD accuracy of the TESCO method in the SECOND dataset. Unlike other datasets, the performance of TESCO on the SECOND dataset is unsatisfactory, and the change detection accuracy is only slightly improved when added to the ChangeStar and FarSeg methods. This is because the SECOND dataset does not have semantic labels in the unchanged areas, resulting in the inability to supervise the multitemporal semantic features of the unchanged areas after TSFB. This shows that semantic supervision of unchanged areas is very important for the TESCO module.

E. Influence of Imbalance Between Changed and Unchanged Samples

The results on the SECOND dataset lead to the following questions: 1) the number of labeled unchanged areas that make the TESCO module effective and 2) the impact of sample imbalance between changed and unchanged areas. In order to answer the above questions, we conducted experiments on the Hi-UCD dataset to verify the impact of different proportions of changed and unchanged samples. In the Hi-UCD training set, the changed samples are about 1/70 of the unchanged samples, and in the test set, it is 1/50. We filtered all the unchanged patches in the Hi-UCD training set (including 12 000 pairs of image patches), obtaining only 1117 pairs with changed samples. The changed pixels accounted for approximately 12% of all 1117 pairs of image patches. The 1117 changed pairs of the Hi-UCD training set were used for the training. We supervised a randomly selected sample of unchanged areas, the number of which was proportional to the number of changed samples. FarSeg and ChangeStar were chosen as the benchmark methods to study the impact of different changed and unchanged sample proportions. The number of training iterations was 15 000, and the evaluation was conducted on the whole Hi-UCD test set. The other settings were the same as in the previous experiments.

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TABLE XV
QUALITATIVE ASSESSMENT RESULTS ON THE HI-UCD TEST SET OF MULTITASK METHOD WITH/WITHOUT THE TESCO MODULE, TRAINED BY DIFFERENT SAMPLE RATIOS

Method	Semantic segmentation			Generation of change	Semantic change detection			Binary change detection	$mIoU_{scc}$
	$mIoU_{seg}(\%)$	$mIoU^{2018}_{seg}(\%)$	$mIoU^{2019}_{seg}(\%)$		$mIoU_{sc}(\%)$	$mIoU^{c}_{sc}(\%)$	$mIoU^{nc}_{sc}(\%)$	$mIoU_{bc}(\%)$	
Changed: Unchanged = 1:0.5									
ChangeStar [9]	**51.56**	50.07	53.05	MASK	**30.05**	13.37	47.72	**26.44**	**45.65**
				PCC	**26.22**	7.53	44.92	**7.57**	
+ TESCO	**50.35**	49.85	50.85	MASK	**29.07**	12.03	46.11	**24.32**	**76.40**
				PCC	**28.15**	10.59	45.71	**19.34**	
Changed: Unchanged = 1:1									
ChangeStar [9]	**50.12**	48.41	51.82	MASK	**28.85**	13.13	44.57	**26.04**	**50.05**
				PCC	**25.14**	7.49	42.79	**7.19**	
+ TESCO	**51.44**	50.78	52.10	MASK	**30.69**	13.25	48.13	**28.76**	**75.59**
				PCC	**29.46**	11.35	47.58	**22.54**	
Changed: Unchanged = 1:2									
ChangeStar [9]	**51.70**	50.47	52.94	MASK	**30.17**	13.08	47.26	**28.07**	**50.12**
				PCC	**26.54**	7.82	45.25	**7.51**	
+ TESCO	**52.75**	52.10	53.40	MASK	**31.06**	13.53	48.59	**27.57**	**76.26**
				PCC	**30.14**	12.01	48.25	**21.38**	
Changed: Unchanged = 1:4									
ChangeStar [9]	**51.06**	50.06	52.05	MASK	**30.11**	14.18	46.03	**27.82**	**45.53**
				PCC	**25.92**	7.76	44.08	**7.39**	
+ TESCO	**54.07**	53.88	54.26	MASK	**32.56**	14.43	50.68	**29.32**	**78.26**
				PCC	**31.40**	12.48	50.32	**25.48**	
Changed: Unchanged ≈ 1:9									
ChangeStar [9]	**50.50**	48.39	52.61	MASK	**29.18**	13.45	44.91	**27.49**	**49.26**
				PCC	**25.37**	7.65	43.10	**7.17**	
+ TESCO	**55.00**	54.61	55.39	MASK	**32.52**	13.68	51.38	**29.15**	**79.33**
				PCC	**31.77**	12.54	51.01	**25.10**	
Changed: Unchanged ≈ 1:70									
ChangeStar [9]	**56.31**	55.56	57.05	MASK	**31.92**	12.73	51.12	**27.67**	**46.63**
				PCC	**29.13**	8.99	49.27	**9.23**	
+ TESCO	**57.60**	57.57	57.62	MASK	**33.41**	12.99	53.84	**26.83**	**77.31**
				PCC	**34.28**	14.53	54.04	**28.84**	

The PCC method is the only way to detect change when only changed samples are available. In Table XIV, the TESCO module effectively improves the PCC method's performance, even with a limited number of unchanged samples. Overall, the greater the number of unchanged samples, the higher the mIoU score, but the gain from the TESCO module decreases, and finally stabilizes at around 3.7% when the number of change sample is 1/9 unchanged sample. When the unchanged samples are further increased and a state of extreme imbalance is reached (changed: unchanged ≈ 1:70), the SCD accuracy still improves by nearly 5%.

As shown in Table XV, the TESCO module still significantly improves the semantic change consistency ($mIoU_{scc}$), regardless of whether there are more or fewer changed samples. However, the reduction of unchanged samples reduces the accuracy of the results generated using PCC, and its score cannot exceed that of the MASK method. Therefore, in the case of fewer unchanged samples, we prefer to use the MASK method to generate the final semantic change results.

F. Impact on the Similarity of Semantic Features in Changed Areas

As discussed in Section III-B1, the ideal value of W_b in the changed area is 1 when using bidirectional semantic feature blending, which can keep the feature distance of multitemporal semantic features in changed areas. The accuracy of W_b for changed and unchanged areas is highly affected by temporal fusion. Since the spatiotemporal features obtained by temporal fusion are difficult to correctly highlight changing areas when encountering radiometric differences, geometric errors, sample imbalances, etc. [10], affecting the estimation of W_b, this may reduce the semantic feature discriminability of changed regions.

In order to further explore the impact of the TESCO module on the feature similarity of changing regions, we used cosine

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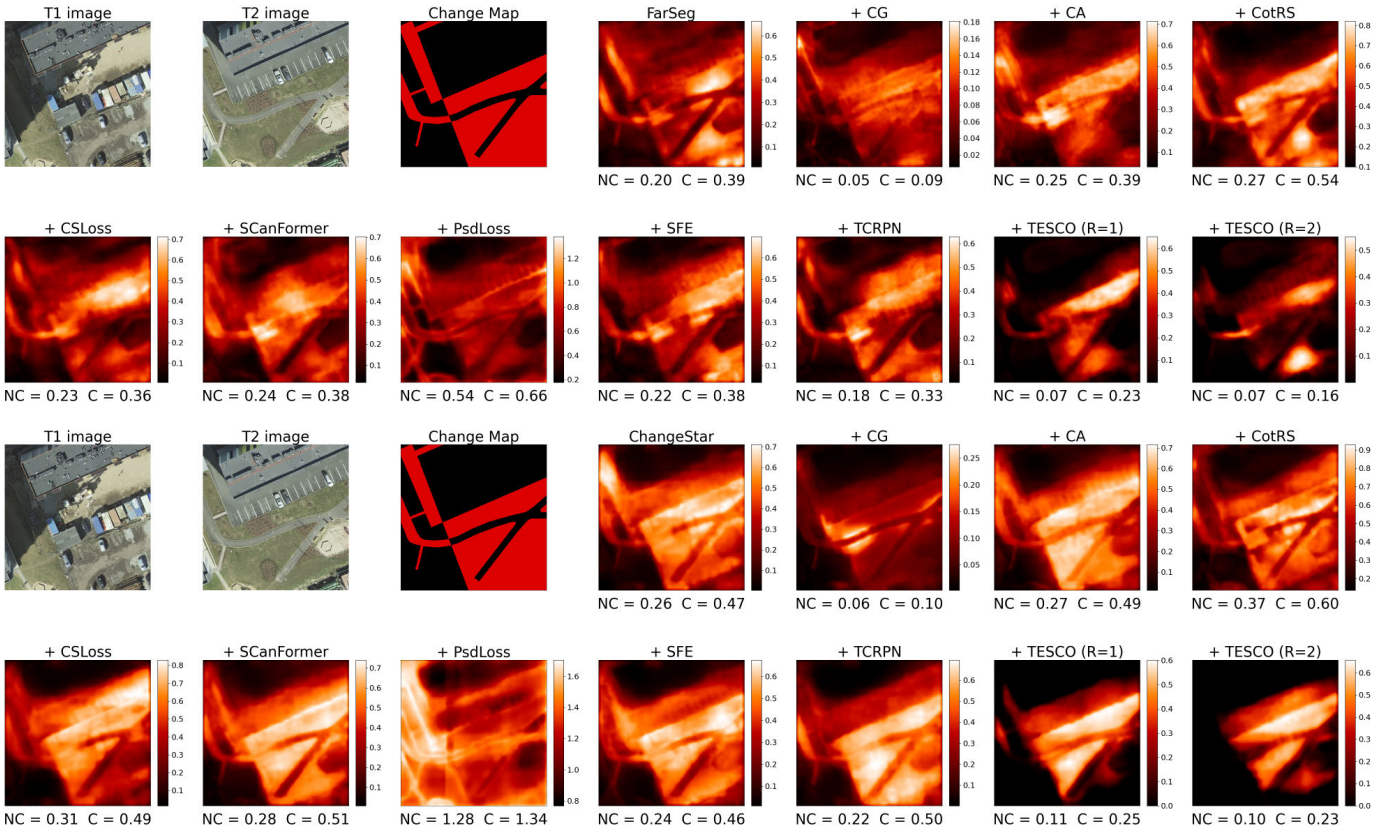


Fig. 15. Similarity of temporal semantic features in different methods. The average similarity of all pixels in the changed area and the unchanged area are recorded, where NC represents the unchanged area and C represents the changed area. The higher the value, the lower the similarity.

similarity (ρ) to calculate the similarity of multitemporal semantic features in different methods on Hi-UCD test set, and the results ($1 - \rho$) are visualized in Fig. 15. At the same time, we calculated the average similarity of all pixels in the changed area and the unchanged area, recorded under each picture (where NC represents the unchanged area and C represents the changed area). As shown in Fig. 15, after using the TESCO module, regardless of the supervision strategy adopted, the similarity between the semantic features of some changed regions is improved. Comparing the average similarity of different methods between the entire changed area and the unchanged area, TESCO expands the ratio of similarity between the changed area and the unchanged area from less than 2 to greater than 2, thereby improving the distinguishability between the changed and unchanged areas, making the change can be correctly identified. Therefore, the TESCO module can overcome the uncertainty generated in W_b estimation.

VI. Conclusion

Temporal consistency is a vital characteristic of multitemporal imagery for SCD, but the existing methods cannot automatically learn this information to produce semantic consistency in unchanged areas. In this study, we explicitly modeled temporal consistency between multitemporal semantic labels and extended it to multitemporal semantic features. Based on the expression of the temporal consistency at features, we proposed TSFB, which can minimize the feature distance to maintain semantic consistency in the unchanged areas by controlling the blending weight. To embed TSFB in SCD networks and realize end-to-end training, we proposed the TESCO module to make the SCD network learn temporal consistency. This plug-and-play module can automatically estimate the blending weight of TSFB and embed temporal correlation into the multitemporal semantic features from coarse to fine by recurrence. To demonstrate the effectiveness of the TESCO module, we added it to several representative SCD methods and conducted comprehensive experiments. The extensive experimental analysis showed that the TESCO module can effectively improve the performance of PCC and obtain competitive results, with better semantic change consistency in multitask SCD methods. Analyzing the experimental results obtained on the three datasets, we noticed some missed detections in the results of methods with the TESCO module, which was because the spatiotemporal features did not highlight the accurate area of change, thereby estimating the wrong blending weight. Foundation models in remote sensing, such as SatMAE [45] and SpectralGPT [46], have been demonstrated to possess powerful spatiotemporal feature extraction capabilities and are widely applied in the BCD task. In the future, we will endeavor to integrate the foundation model in order to more precisely highlight the change areas and enhance TESCO's capacity to detect semantic changes in time-series imagery. We hope that the TESCO module will serve as a solid baseline and provide new inspiration to improve the performance of SCD methods.

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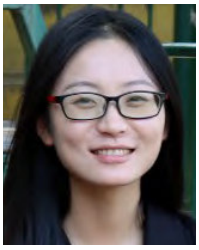
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